A Project Report

on

SECURITY ANALYSIS OF AUTONOMOUS CYBER-PHYSICAL SYSTEMS USING DEEP LEARNING

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY
IN
INFORMATION TECHNOLOGY

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April 2026

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It is further certified that this work has not been submitted, either in part of in full, to any other department of the Vasireddy Venkatadri Institute of Technology, or any other University, institution or elsewhere, or for publication in any form.

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ACKNOWLEDGEMENT

We take this opportunity to express our deepest gratitude and appreciation to all those people who made this project work easier with words of encouragement, motivation, discipline, and faith by offering different places to look to expand my ideas and helped me towards the successful completion of this project work.

First and foremost, we express our deep gratitude to **Sri. Vasireddy Vidya Sagar, Chairman,** Vasireddy Venkatadri Institute of Technology for providing necessary facilities throughout the Information Technology program.

We express our sincere thanks to **Dr. Y. Mallikarjuna Reddy, Principal,** Vasireddy Venkatadri Institute of Technology for his constant support and cooperation throughout the Information Technology program.

We express our sincere gratitude to **Dr. A. Kalavathi, Professor & HOD,** Information Technology, Vasireddy Venkatadri Institute of Technology for her constant encouragement, motivation and faith by offering different places to look to expand my ideas. We would like to express our sincere gratefulness to our guide **Dr. K. Kranthi Kumar, Professor** for his insightful advice, motivating suggestions, invaluable guidance, help and support in successful completion of this project and also our project coordinator **Dr. K. Kranthi Kumar, Professor** for his advice and support. We would like to take this opportunity to express our thanks to the teaching and non-teaching staff in Department of Information Technology, VVIT for their invaluable help and support.

ABSTRACT

Cyber-Physical Systems (CPS) are a combination of physical processes and computational technologies that enable the intelligent monitoring and control of industrial environments through the Industrial Internet of Things (IIoT). This combination is being used in numerous industries, including smart manufacturing, energy management, and automated control systems; however, due to their interconnectedness with devices within the network and their integrated physical infrastructure, CPS are vulnerable to several types of cyberattacks, including Distributed Denial of Service (DDoS), Denial of Service (DoS), Man-In-The-Middle (MITM), and injection attacks. The complexity and high dimensionality of network traffic data generated by CPS networks present challenges for detecting cyber attacks.This study proposes a deep-learning-based intrusion detection framework for CPS-type environments. The architecture utilizes both Convolutional Neural Networks (CNNs) and Long-Short Term Memory Networks (LSTMs) to capture both feature dependencies and temporal communication patterns in network traffic. In order to increase the model's robustness, Generative Adversarial Networks (GANs) will create additional synthetic samples during the training process. Additionally, the explainable AI techniques will be used to analyze the model's behavior through layer-wise analysis and identify the most significant features in the classification of attacks. Using the Edge-IIoT Balanced Subset for Intrusion Detection dataset, which contains 50,000 network traffic samples, the proposed architecture correctly detects cyberattacks in multiple different categories and provides explainable information about CPS security.

Keywords: Cyber-Physical Systems, Industrial Internet of Things, Intrusion Detection System, Deep Learning, CNN-LSTM, Generative Adversarial Networks, Explainable Artificial Intelligence, Network Traffic Analysis, Attack Detection, Edge-IIoT Dataset.

TABLE OF CONTENTS

ABSTRACT		v
LIST OF TABLES		viii
LIST OF FIGURES		ix
LIST OF ABBREVIATIONS		x
CHAPTER 1	INTRODUCTION	11
1.1	ABOUT THE PROJECT	11
1.1.1	PROPOSED SYSTEM	14
1.2	SOFTWARE REQUIREMENTS	14
1.3	HARDWARE REQUIREMENTS	15
CHAPTER 2	LITERATURE SURVEY	17
CHAPTER 3	SYSTEM ANALYSIS	20
3.1	INTRODUCTION	20
3.2	FEASIBILITY STUDY	20
3.3	FUNCTIONAL REQUIREMENTS	22
3.4	NON-FUNCTIONAL REQUIREMENTS	22
CHAPTER 4	SYSTEM DESIGN	23
4.1	INTRODUCTION TO SYSTEM DESIGN	23
4.2	SYSTEM ARCHITECTURE	24
4.3	USE CASE DIAGRAM	27
4.4	CLASS DIAGRAM	28
4.5	ACTIVITY DIAGRAM	29
4.6	SEQUENCE DIAGRAM	30
4.7	DEPLOYMENT DIAGRAM	31

CHAPTER 5	SYSTEM IMPLEMENTATION	32
5.1	INTRODUCTION TO PYTHON	32
5.2	TOOLS AND TECHNOLOGIES USED	33
5.3	ALGORITHM DETAILS	35
CHAPTER 6	SYSTEM TESTING	42
6.1	INTRODUCTION TO SYSTEM TESTING	42
6.2	FUNCTIONAL TESTING	42
6.3	NON - FUNCTIONAL TESTING	43
6.4	TESTING AND RESULTS	44
6.5	PERFORMANCE TESTING	45
6.6	RESULTS AND OUTPUT SCREENS	46
6.7	DISCUSSIONS	51
6.8	SUMMARY	51
CHAPTER 7	CONCLUSION AND FUTURE SCOPE	52
7.1	CONCLUSION	52
7.2	FUTURE SCOPE	52
REFERENCES		54

LIST OF TABLES

TABLES	PAGE NO
Table 5.3.1: Selected Input Features and its Description	35
Table 6.4: Testing and Results	44

LIST OF FIGURES

FIGURES	PAGE NO
Figure 1: System Architecture of CPS Intrusion Detection System	25
Figure 2: Use Case Diagram of CPS Intrusion Detection	27
Figure 3: Class Diagram of CPS Intrusion Detection	28
Figure 4: Activity Diagram of CPS Intrusion Detection	29
Figure 5: Sequence Diagram of CPS Intrusion Detection	30
Figure 6: Deployment Diagram of CPS Intrusion Detection	31
Figure 7: Displayed CPS Intrusion Detection System Dashboard Interface	46
Figure 8: Displayed Model Training Results and Performance Metrics	46
Figure 9: Real-Time Detection Input Interface for CPS Network Traffic	47
Figure 10: Displayed Intrusion Detection Output with XAI, Feature Importance	48
Figure 11: Accuracy Graph	49
Figure 12: Loss Graph	49
Figure 13: Confusion Matrix	50

LIST OF ABBREVIATIONS

CPS	Cyber-Physical Systems
IIoT	Industrial Internet of Things
DDoS	Distributed Denial of Service
DoS	Denial of Service
MITM	Man-In-The-Middle
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
GAN	Generative Adversarial Network
XAI	Explainable Artificial Intelligence
IDS	Intrusion Detection System
AI	Artificial Intelligence
CSV	Comma Separated Values

CHAPTER 1
INTRODUCTION

1.1 ABOUT THE PROJECT

The rapid expansion of intelligent industrial infrastructure has increased the adoption of Cyber-Physical Systems (CPS) in modern applications. These applications include smart manufacturing, intelligent transportation, smart grids, and industrial automation. CPS integrates computational intelligence, communication networks, and physical processes for real-time monitoring. Through the Industrial Internet of Things (IIoT), these systems exchange large volumes of operational data. This integration significantly improves efficiency, productivity, and automated control in physical environments. However, the growing level of connectivity also introduces serious cybersecurity challenges.

Traditional intrusion detection systems mainly depend on signature-based and rule-based detection mechanisms. Such approaches are often ineffective in identifying newly evolved and sophisticated cyber-attacks. CPS networks generate high-dimensional and large-scale traffic data that is difficult to analyze. Conventional machine learning techniques struggle to capture complex patterns across multiple interconnected networks. As a result, detecting multi-stage and coordinated attacks becomes more challenging. Deep learning methods have recently shown strong potential in overcoming these cybersecurity limitations.

To address this issue, a hybrid CNN-LSTM-based intrusion detection system is proposed. An enhanced version of the model incorporates GAN-based data augmentation techniques. This approach improves learning capability when dealing with limited or imbalanced datasets. A hybrid Explainable AI (XAI) framework is also integrated to provide interpretable results. The models are evaluated using the Edge-IIoT Balanced Subset for Intrusion Detection dataset containing 50,000 samples. The objective is to improve intrusion detection accuracy while identifying key features affecting CPS network security.

Problem Statement

Despite the advancements in CPS and IIoT technologies, ensuring robust cybersecurity remains a major challenge. The key problems identified in existing systems are:

- **High Complexity of CPS Networks:** CPS generates large-scale and high-dimensional data, making it difficult to analyze using traditional methods.
- **Inefficient Detection of Advanced Attacks:** Conventional intrusion detection systems are unable to detect sophisticated and multi-stage cyberattacks.
- **Lack of Real-Time Detection:** Many existing systems fail to provide real-time monitoring and quick response to cyber threats.
- **Imbalanced Datasets:** Cybersecurity datasets often contain fewer attack samples, leading to poor model performance.
- **Limited Interpretability:** Most deep learning models act as black boxes, making it difficult to understand their decision-making process.
- **Poor Handling of Temporal Patterns:** Traditional machine learning models fail to capture time-based attack behaviors in network traffic.

Due to these limitations, there is a critical need to design a **robust, intelligent, and explainable intrusion detection system** that can effectively secure CPS environments.

Objectives

The main objective of this project is to develop an efficient and intelligent intrusion detection system for securing Cyber-Physical Systems (CPS) using advanced deep learning techniques. The system is designed to address the growing security challenges in Industrial Internet of Things (IIoT) environments by providing accurate, reliable, and real-time detection of cyber threats.

This project focuses on designing a hybrid deep learning model that combines Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks to effectively capture both spatial and temporal characteristics of network traffic data. The CNN component is responsible for extracting meaningful spatial features and identifying hidden patterns, while the LSTM component analyzes sequential dependencies to detect time-based and multi-stage cyberattacks.

To further enhance the performance of the model, Generative Adversarial Networks (GAN) are utilized for data augmentation. This helps in generating synthetic samples and addressing the issue of imbalanced datasets, thereby improving the model's ability to detect rare and zero-day attacks. Additionally, Explainable Artificial Intelligence (XAI) techniques are integrated into the

system to provide transparency and interpretability of the model's decisions, enabling better understanding of how different features influence attack classification.

Another important objective of this project is to develop a real-time intrusion detection system capable of continuously monitoring CPS network traffic and classifying it into normal or malicious categories. The system is designed to detect multiple types of cyberattacks, including Distributed Denial of Service (DDoS), Denial of Service (DoS), Man-in-the-Middle (MITM), and injection attacks, ensuring comprehensive security coverage.

Furthermore, the system aims to minimize false positives and false negatives to improve detection accuracy and reliability. It is also designed to be scalable so that it can efficiently handle large-scale CPS environments with high volumes of network traffic. Emphasis is given to enhancing overall system performance and stability, ensuring consistent operation under different conditions.

In addition to detection capabilities, the project also focuses on providing visualization and analytical insights into the results, enabling better evaluation of system performance and aiding in decision-making processes. Overall, the objective is to build a robust, scalable, and intelligent cybersecurity framework for modern CPS environments.

The significance of this project lies in improving the security of Cyber-Physical Systems used in Industrial Internet of Things environments. With the increasing use of CPS in industries such as manufacturing and energy systems, protecting these systems from cyber threats has become very important.

The proposed system uses deep learning techniques to provide accurate and efficient intrusion detection. The hybrid CNN-LSTM model helps in identifying both spatial and temporal patterns in network traffic, enabling detection of complex cyberattacks. The use of GAN improves the system's ability to handle imbalanced datasets and detect rare attacks.

In addition, the integration of Explainable Artificial Intelligence enhances transparency by providing insights into model decisions. The system also supports real-time monitoring, making it useful for practical applications.

Bottom of Form

1.1.1 PROPOSED SYSTEM

The proposed system integrates real-time network monitoring capabilities for securing Cyber-Physical Systems in Industrial Internet of Things environments. It utilizes a hybrid CNN-LSTM deep learning architecture to capture spatial traffic features and temporal communication patterns. The CNN component extracts protocol-level characteristics and abnormal network behavior related to cyber threats. The LSTM component analyzes sequential traffic patterns to detect coordinated and multi-stage cyberattacks. GAN-based data augmentation generates realistic synthetic samples to address dataset imbalance and improve detection robustness. Explainable Artificial Intelligence techniques help interpret model decisions by identifying important features influencing attack classification. Continuous model refinement and parallel processing enable adaptive performance and efficient real-time intrusion detection.:

- **Real-time Intrusion Detection Capability:** Develop a system capable of monitoring CPS network traffic in real time and detecting cyber-attacks with minimal latency.
- **Hybrid Spatial–Temporal Feature Learning:**.Utilize CNN layers to extract protocol-level traffic features and LSTM layers to capture sequential attack behaviors for improved detection accuracy.
- **Improved Detection of Rare and Zero-day Attacks:** Apply GAN-based data augmentation to generate synthetic attack samples and balance the dataset for robust learning.
- **Enhanced Model Interpretability:** Integrate Explainable AI techniques to identify influential features and provide transparency in attack classification decisions.
- **Adaptive and Scalable Security Framework:** Design the system to support dynamic CPS environment and large-scale IIoT networks with high dimensional traffic data.

1.2 SOFTWARE REQUIREMENTS

- **Python Libraries**: TensorFlow / PyTorch, Keras, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, Gradient-based Explainable AI (XAI), Layer-wise Activation Analysis.

- **Python:** The proposed intrusion detection system is implemented using Python, so Python must be installed on the system. It is recommended to use the latest stable version such as Python 3.9 or above for better compatibility with deep learning frameworks.
- **Deep Learning Libraries:** TensorFlow or PyTorch is used for building and training the hybrid CNN-LSTM model and GAN-based data augmentation framework.
- **Data Processing Libraries:** NumPy and Pandas are used for preprocessing CPS network traffic data, feature handling, and dataset manipulation.
- **Machine Learning Library:** Scikit-learn is used for dataset splitting, normalization, performance evaluation metrics, and confusion matrix generation.
- **Visualization Libraries:** Matplotlib and Seaborn are used for plotting training accuracy, loss graphs, and performance comparison charts.
- **Explainable AI Library:** Gradient-based explainability techniques using TensorFlow's GradientTape are employed to interpret model predictions by computing feature importance based on the sensitivity of the output with respect to input features.
- **Dataset Requirement:** Edge-IIoT Balanced Subset for Intrusion Detection dataset is required for training and evaluating the intrusion detection model.
- **Development Environment (IDE):** Visual Studio Code, PyCharm, or Jupyter Notebook can be used for coding, execution, and debugging.

1.3 HARDWARE REQUIREMENTS

- **CPU:**
  - Minimum: Quad-core processor clocked at 2.5 GHz
  - Maximum: Octa-core processor clocked at 3.5 GHz or higher
- **RAM:**
  - Minimum: 8GB DDR4 RAM
  - Maximum: 32GB DDR4 RAM or more for efficient deep learning training
- **STORAGE:**
  - Minimum: SSD storage with at least 256GB capacity

- Maximum: SSD with 1TB or higher for faster dataset access and model storage

- **GPU:**
  - Minimum: Integrated GPU or entry-level dedicated GPU
  - Maximum: NVIDIA mid to high-end GPU with CUDA support for accelerated deep learning model training

CHAPTER 2
LITERATURE SURVEY

The integration of deep learning with cyber-physical systems (CPS) has been effectively applied in smart healthcare, particularly for brain CT image interpretation. This approach enhances diagnostic accuracy and enables automated decision-making by processing real-time medical data efficiently [1].

The combination of deep learning and reinforcement learning has been explored to improve adaptability in autonomous mobile CPS. Such systems can learn from dynamic environments and optimize their decisions, making them suitable for real-time applications [2].

A hybrid approach using reinforcement learning and supervised learning has been proposed for fault detection and tolerance in CPS. This method improves system reliability by identifying faults early and enabling corrective actions [3].

Artificial intelligence techniques have been widely used to enhance cyber-physical security systems. These methods support automated monitoring, anomaly detection, and improved response to cyber threats [4].

Security and privacy challenges in CPS, such as data breaches and system vulnerabilities, have been extensively studied. Advanced solutions including encryption and AI-based protection mechanisms have been suggested to address these issues [5].

Automated knowledge-based cybersecurity risk assessment models have been developed to evaluate threats in CPS. These models help identify vulnerabilities and prioritize appropriate security measures [6].

Risk assessment in autonomous CPS has been emphasized with a focus on safety and reliability. Integrating security measures into system design is crucial for preventing cyber threats [7].

Deep learning-based intrusion detection systems have been widely studied, highlighting their effectiveness in detecting cyber attacks. However, challenges such as data imbalance and scalability remain significant [8].

Comparative studies between production systems and cyber-physical systems provide insights into their architectures and functionalities, helping in understanding the evolution of CPS technologies [9].

Zero Trust architecture has been proposed for industrial CPS to ensure continuous authentication and minimize unauthorized access. This approach strengthens overall system security [10].

Current deep learning-based intrusion detection models face challenges such as interpretability and handling large-scale data. Future improvements are required to enhance their performance [11].

Reinforcement learning has been applied to improve cybersecurity in autonomous vehicles, enabling adaptive responses to threats and enhancing system safety [12].

Federated learning has been introduced for CPS security, allowing decentralized model training while preserving data privacy. This approach is beneficial for distributed environments [13].

Machine learning techniques have been used for intelligent vulnerability detection in CPS, helping identify weaknesses and improve system robustness [14].

Multi-agent systems using Double Deep Q-Learning have been proposed for autonomous decision-making in CPS. These systems improve coordination and efficiency in complex environments [15].

Learning-based recovery mechanisms have been developed to restore CPS after sensor attacks. These approaches enhance system resilience and ensure continuous operation [16].

Generative AI and cognitive computing have been utilized to design advanced intrusion detection systems. These systems provide improved accuracy and adaptability against evolving cyber threats [17].

Learning-based vulnerability analysis techniques help identify system weaknesses and improve CPS security through proactive measures [18].

Deep reinforcement learning has shown great potential in cybersecurity applications by enabling proactive and adaptive defense strategies [19].

Intrusion detection systems have been specifically designed for unmanned aerial vehicles (UAVs) to detect anomalies in communication and control systems, improving UAV security [20].

Verification techniques for learning-enabled CPS ensure the reliability and correctness of AI-based decision-making processes, which is critical for safety-critical systems [21].

TinyML-based approaches have been proposed to enhance transportation CPS security by enabling lightweight and efficient machine learning models on edge devices [22].

Meta-learning techniques have been applied to improve autonomous CPS navigation and adaptability, enabling better performance in dynamic environments [23].

Cyber-physical security analytics have been developed for smart grid systems to detect anomalies and ensure secure energy transactions [24].

Variational autoencoders have been used to detect adversarial attacks in CPS, improving the robustness of learning-enabled systems [25].

Deep learning-enabled intrusion detection systems have been proposed for IoT environments, providing improved detection accuracy and security [26].

Recent studies highlight challenges in deep learning-based IDS, such as feature extraction and data imbalance, and suggest advanced techniques to overcome them [27].

AI-driven intrusion detection and prevention systems have been developed for 6G networks, ensuring protection against advanced cyber threats [28].

Novel deep learning-based intrusion detection systems for Industry 4.0 have demonstrated improved performance in detecting cyber attacks in modern industrial environments [29].

CHAPTER 3

SYSTEM ANALYSIS

3.1 INTRODUCTION

In our research, securing Autonomous Cyber-Physical Systems (CPS) is critical due to the increasing deployment of Industrial Internet of Things technologies in smart industries. Existing intrusion detection approaches struggle to effectively identify sophisticated cyberattacks in high-dimensional CPS network traffic environments. The proposed system introduces a deep learning-based framework for real-time intrusion detection and security monitoring. By leveraging hybrid CNN-LSTM architecture, the system captures both spatial network traffic characteristics and temporal attack behaviors. GAN-based data augmentation enhances the detection capability by generating realistic synthetic attack samples. Explainable Artificial Intelligence techniques further improve transparency by identifying important features influencing cyberattack classification.

3.2 FEASIBILITY STUDY

The feasibility study of this project has revealed the project as follows:

3.2.1 Technical Feasibility:

- The system utilizes deep learning frameworks such as TensorFlow or PyTorch for building CNN-LSTM and GAN models, which are technically feasible with modern computing resources.
- Real-time intrusion detection is achieved through optimized data preprocessing and model inference mechanisms.
- Dataset balancing and synthetic attack generation are implemented using GAN models to improve robustness.
- Model evaluation tools and visualization libraries ensure effective performance monitoring and analysis.

3.2.2 Economic Feasibility:

- The The system relies mainly on open-source tools and datasets such as Python libraries and the Edge-IIoT Balanced Subset for Intrusion Detection dataset, reducing licensing costs.
- Scalable architecture allows deployment in different CPS environments without major

infrastructure investment.

- Efficient resource utilization during training and inference minimizes the need for expensive hardware upgrades.
- Modular framework design supports easy integration and reduces long-term maintenance expenses.

3.2.3 Operational Feasibility:

- The The deep learning-based detection framework supports continuous monitoring of CPS network traffic for improved operational security.
- Automated feature extraction and classification reduce manual intervention in cybersecurity monitoring.
- Real-time detection capability enhances system responsiveness to cyber threats.
- Flexible architecture allows adaptation to different industrial environments and network configurations

3.2.4 Schedule Feasibility:

- Leveraging Availability of benchmark datasets and open-source deep learning frameworks accelerates system development.
- Modular design enables parallel implementation of preprocessing, model training, and evaluation components.
- Pretrained architectures and optimization techniques reduce overall development time.
- Efficient experimentation and performance tuning help in meeting project timelines.

3.2.5 Ethical Feasibility:

- Ensure Ensure responsible use of cybersecurity data and maintain confidentiality of network traffic information.
- void bias in model predictions by using balanced datasets and proper validation strategies.
- Maintain transparency in decision-making using Explainable AI techniques.Follow ethical AI practices and data protection guidelines during system deployment.

3.3 FUNCTIONAL REQUIREMENTS

- **Network Traffic Acquisition:** The system should collect CPS network traffic data for analysis.
- **Data Preprocessing:** Normalize and transform raw traffic data for deep learning model input.
- **Intrusion Detection:** Classify network traffic into normal or attack categories using CNN-LSTM models.
- **Data Augmentation:** Generate synthetic attack samples using GAN to improve model training.
- **Text Overlay:** Display the extracted and translated text on video frames.
- **Real-time Monitoring:** Provide continuous detection and alert mechanisms for cyber threats.
- **Frame Storage:** Save the extracted frames as PNG images in a designated directory.
- **Result Storage:** Save classification results, performance metrics, and logs for further analysis.

3.4 NON-FUNCTIONAL REQUIREMENTS

- **Performance:** The system should ensure low latency and high accuracy in real-time intrusion detection.
- **Usability:** The framework should be easy to deploy and configure in industrial CPS environments.
- **Reliability:** The detection model should perform consistently across different network conditions.
- **Scalability:** The architecture should support large-scale IIoT network traffic analysis.
- **Maintainability:** The Modular and well-documented code should support future enhancements.
- **Security:** Ensure secure handling of datasets, model outputs, and system configurations.

CHAPTER 4
SYSTEM DESIGN

4.1 INTRODUCTION OF SYSTEM DESIGN

System design is a crucial stage in developing a deep learning-based intrusion detection system for Cyber-Physical Systems (CPS), where architecture, components, and workflow are defined for accurate threat detection. It focuses on integrating data processing, hybrid deep learning models, and real-time monitoring mechanisms to handle complex CPS network traffic efficiently. This design ensures scalability, reliability, and adaptability for detecting evolving cyberattacks in Industrial Internet of Things environments.

This phase involves several key steps:

1. **Requirement gathering:** Identifying system requirements such as real-time intrusion detection, high accuracy, low latency, and support for multiple cyberattack categories. This includes understanding CPS network environments and security challenges.

2. **Architecture design:** Designing the overall system architecture, including data collection modules, preprocessing units, hybrid CNN-LSTM model, GAN-based data augmentation module, and explainable AI components. It also defines how these components interact within the system.

3. **Component design:** Defining individual components such as data preprocessing pipelines, feature extraction layers, deep learning models, and evaluation modules. This step involves selecting appropriate algorithms for intrusion detection and sequential analysis.

4. **Optimization and refinement:** Improving system performance by tuning hyperparameters, optimizing model accuracy, reducing computational complexity, and enhancing real-time detection capability.

5. **Integration and testing:** Integrating all system components into a unified framework and conducting testing to evaluate performance using metrics such as accuracy, precision, recall, and F1-score under different CPS network conditions.

4.2 SYSTEM ARCHITECTURE

The architecture of the proposed system focuses on real-time intrusion detection in Cyber-Physical Systems operating within Industrial Internet of Things environments. It consists of multiple interconnected components including data collection, preprocessing, feature extraction, deep learning-based detection, and result visualization modules. The system begins with capturing CPS network traffic data from sensors and communication networks. This data is then preprocessed and normalized to remove noise and prepare it for model training. The processed data is fed into a hybrid CNN-LSTM model that identifies both spatial and temporal attack patterns. Each component works together to ensure accurate and efficient detection of cyber threats in real time.

Once the data is processed, the intrusion detection module utilizes Convolutional Neural Networks to extract protocol-level features and abnormal traffic patterns. The Long Short-Term Memory network analyzes sequential dependencies to detect coordinated and multi-stage cyberattacks. To improve model performance, a Generative Adversarial Network is used to generate synthetic attack samples and balance the dataset. This enhances the system's ability to detect rare and unseen threats effectively. Additionally, Explainable Artificial Intelligence techniques are applied to interpret model decisions and identify key features influencing attack classification. This ensures transparency and reliability in cybersecurity decision-making.

To support scalability and flexibility, the system architecture follows a modular design where each component operates independently but communicates through defined interfaces. This allows easy integration with existing CPS infrastructure and supports large-scale IIoT environments. The architecture also enables real-time monitoring and alert generation for detected intrusions. Performance evaluation modules assess accuracy, precision, recall, and other metrics to ensure system effectiveness. Overall, the proposed architecture provides a robust, scalable, and intelligent framework for securing CPS networks against evolving cyber threats.

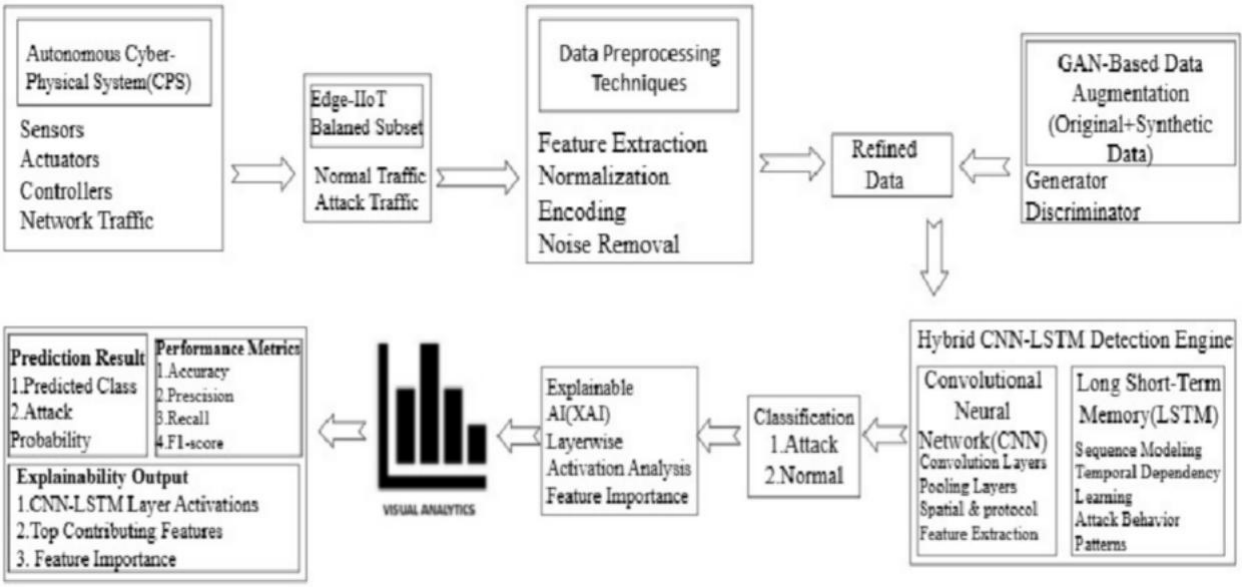


Figure 1: System Architecture of CPS Intrusion Detection System

The System architecture has following components :

- **Input Source (CPS Environment):** The system receives input from CPS including sensors, actuators, controllers, and network traffic generated in Industrial IoT environments.
- **Dataset Collection (Edge-IIoT Balanced Subset for Intrusion Detection Dataset):** Network traffic data consisting of normal and attack traffic (DDoS, DoS, MITM, Injection) is collected for training and testing the intrusion detection system.
- **Pre-processing:** Pre-processing techniques such as normalization, encoding, and noise removal are applied to clean and prepare high-dimensional CPS network traffic data.
- **Feature Extraction:** Important features are extracted from the processed data to represent traffic behavior effectively for deep learning models.
- **GAN-Based Data Augmentation:** Generative Adversarial Networks generate synthetic attack samples to handle dataset imbalance and improve detection of rare and unseen attacks.
- **Hybrid CNN-LSTM Detection Module:** CNN extracts spatial and protocol-level features while LSTM captures temporal dependencies and sequential attack patterns for accurate intrusion detection.
- **Classification Module:** The system classifies network traffic into categories such as normal or attack based on learned patterns from the hybrid model.

- **Explainable AI (XAI) Module:** Provides interpretability by identifying key features and layer-wise activations that influence the model's decision-making process.
- **Performance Evaluation:** Evaluates system performance using metrics such as accuracy, precision, recall, and F1-score to ensure reliable detection.
- **Output (Prediction Results):** Displays predicted class (attack or normal), attack probability, and important contributing features for effective CPS security monitoring.

4.2 USE CASE DIAGRAM

The use case diagram of the CPS intrusion detection system includes three main actors: Security Analyst, Autonomous CPS, and System Admin. The Security Analyst interacts with the system to configure detection parameters, view intrusion alerts, analyze XAI explanations, and evaluate performance metrics. The Autonomous CPS generates network traffic and sends sensor data to the system for analysis. The System Admin is responsible for deploying and maintaining the system. The system processes uploaded Edge-IIoT Balanced Subset for Intrusion Detection dataset, performs preprocessing, applies GAN-based data augmentation, and detects intrusions using the CNN-LSTM model. This interaction ensures effective monitoring and management of CPS network security.

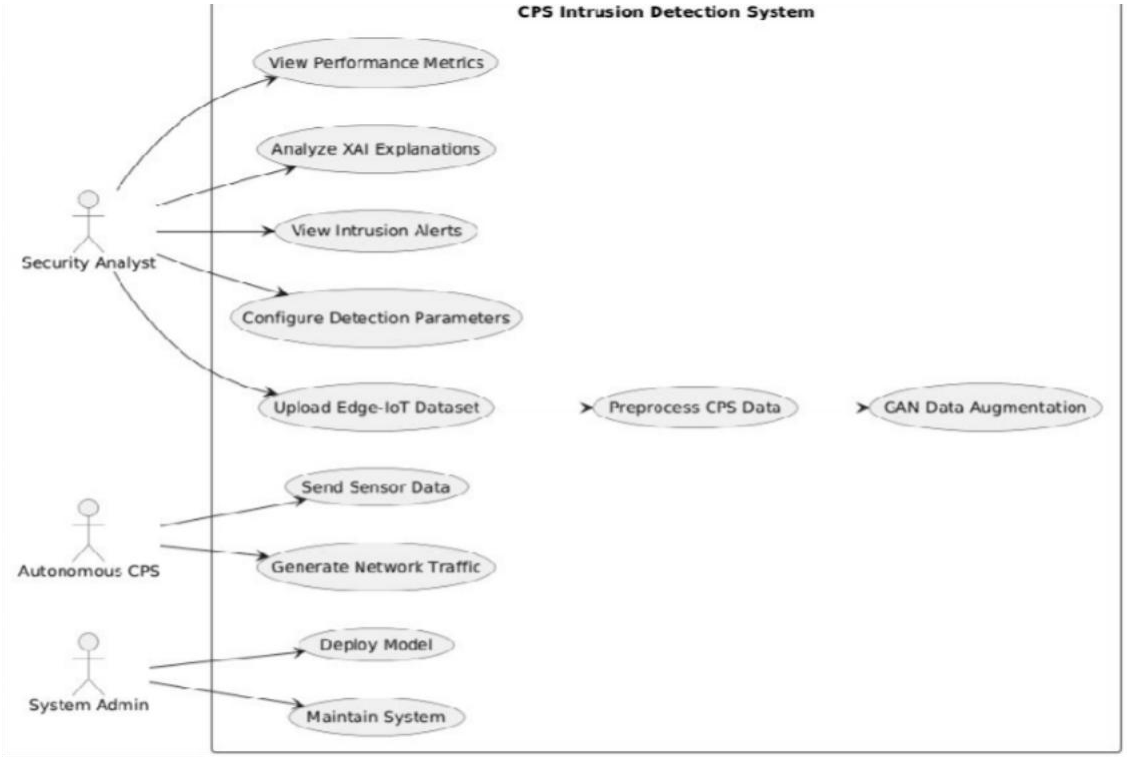


Figure 2: Use Case Diagram of CPS Intrusion Detection System

4.3 CLASS DIAGRAM

The The class diagram defines the core components of the system. The **Dataset** class handles loading, normalization, and splitting of data. The **Preprocessing** class performs data cleaning and encoding operations. The **GAN** class is responsible for training the generator and producing synthetic samples. The **CNN_LSTM** class builds, trains, and predicts intrusion patterns. The **Evaluation** class computes performance metrics such as accuracy and recall. The **XAI** class provides interpretability by explaining model predictions. These classes work together to form a complete intrusion detection framework.

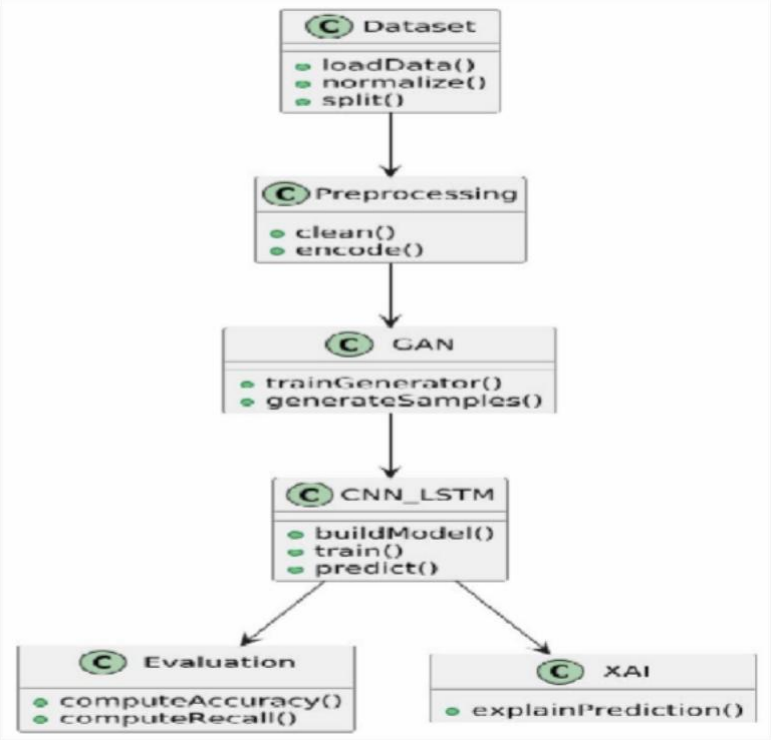


Figure 3: Class Diagram of CPS Intrusion Detection System

4.4 ACTIVITY DIAGRAM

The activity diagram represents the workflow of the intrusion detection system. The process starts when the CPS becomes active and begins data collection from sensors and network traffic. The collected data undergoes preprocessing such as cleaning, normalization, and encoding. If the dataset is imbalanced, GAN-based augmentation is applied to generate synthetic attack samples. The processed data is then used to train the CNN-LSTM model. After training, the system performs intrusion detection on incoming traffic. If an attack is detected, an alert is generated; otherwise, normal traffic is monitored continuously. Finally, the system outputs results and continues monitoring operations..

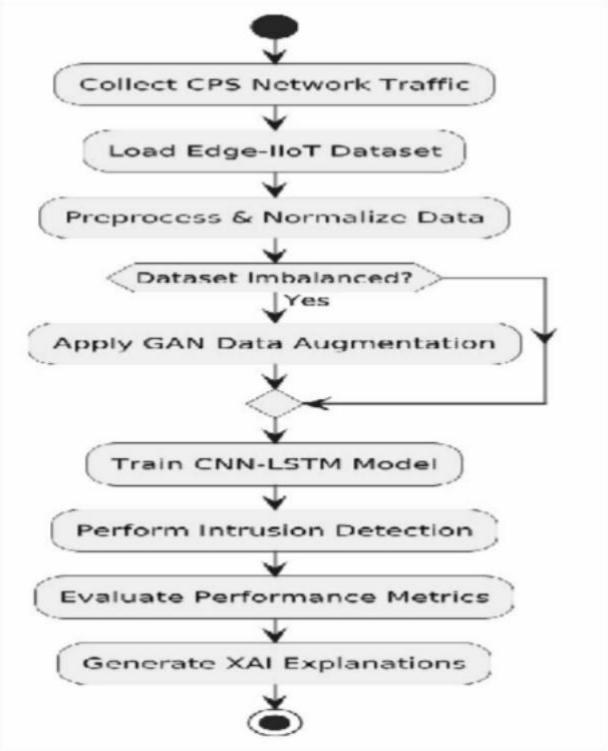


Figure 4: Activity Diagram of CPS Intrusion Detection System

4.5 SEQUENCE DIAGRAM

The The sequence diagram illustrates interactions between the Security Analyst, CPS system, preprocessing module, GAN module, CNN-LSTM model, and evaluation module. Initially, the CPS sends raw network traffic to the preprocessing module. The data is cleaned and normalized before being passed to the GAN module if imbalance is detected. The GAN generates synthetic attack samples and creates an augmented dataset. This dataset is then used to train the CNN-LSTM model, which performs intrusion detection. The model produces detection results, which are evaluated using performance metrics. The system also generates XAI-based explanations and sends both results and explanations back to the analyst.

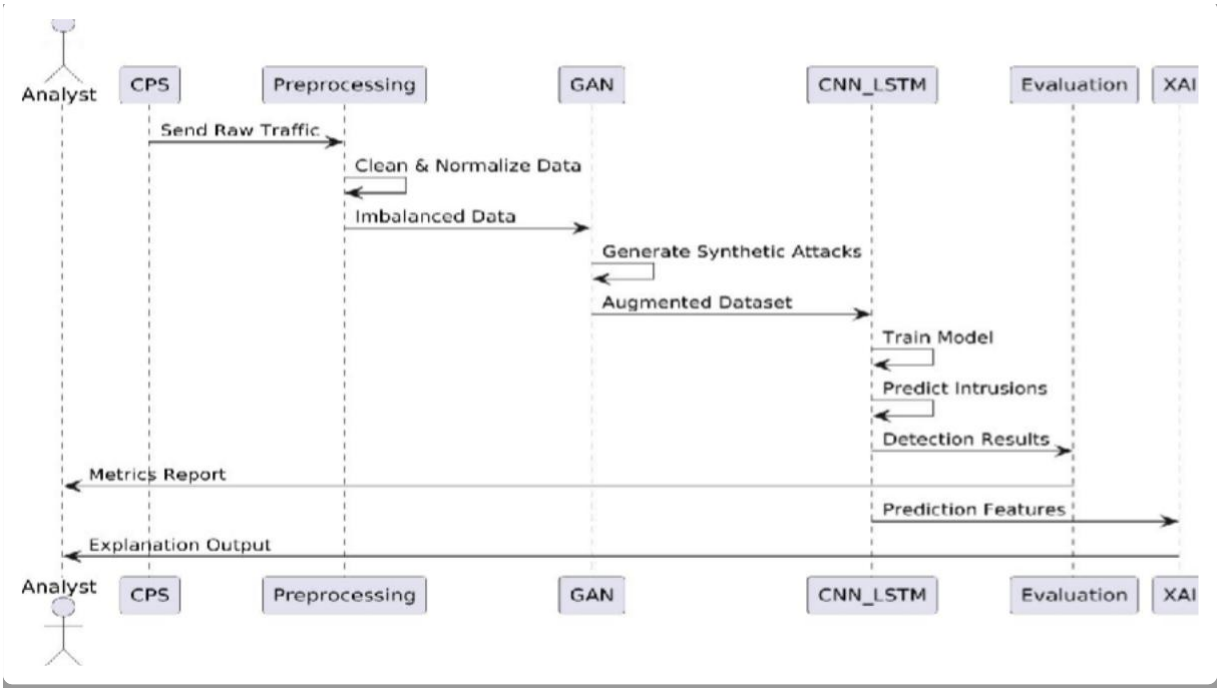


Figure 5: Sequence Diagram of CPS Intrusion Detection System

Deployment Diagram

The Deployment Diagram represents the physical architecture of the proposed intrusion detection system and shows how different software components are deployed on hardware devices. In this system, the user interacts with the application through a frontend interface available on a user device such as a computer or laptop. The main processing of the system, including data preprocessing, GAN-based data augmentation, and the hybrid CNN-LSTM model, is carried out on the application server. This server handles all computational tasks such as training the model, analyzing network traffic, and detecting cyberattacks. The dataset and generated results are stored in a storage system or database, which can be accessed by the application server when required. The communication between these components ensures smooth data flow and efficient system operation. This deployment structure supports scalability, reliability, and real-time intrusion detection in Cyber-Physical System environments.

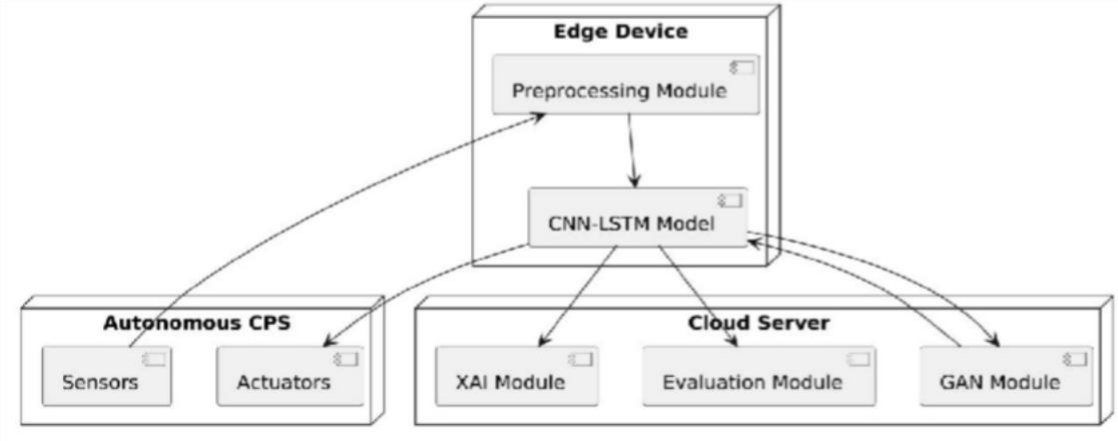


Figure 6: Deployment Diagram of CPS Intrusion Detection

CHAPTER 5
SYSTEM IMPLEMENTATION

5.1 INTRODUCTION TO PYTHON

This This project demonstrates the use of Python for developing a deep learning-based intrusion detection system for Cyber-Physical Systems in Industrial Internet of Things environments. By integrating powerful libraries such as TensorFlow or PyTorch, NumPy, Pandas, and Scikit-learn, the system efficiently processes CPS network traffic data, performs preprocessing, and detects cyberattacks. The hybrid CNN-LSTM model is implemented to capture both spatial and temporal patterns in network traffic for accurate intrusion detection. GAN-based data augmentation is also integrated to generate synthetic samples and improve model robustness. Through modular design and well-structured code, the system ensures readability, scalability, and ease of maintenance. Additionally, proper error handling and validation mechanisms improve system reliability and stability during execution.

Furthermore, the project highlights Python's capability in handling complex cybersecurity tasks such as anomaly detection and attack classification in CPS environments. The integration of deep learning frameworks enables efficient model training and real-time prediction. Parallel processing and optimized data handling techniques improve system performance when dealing with large-scale network traffic datasets. Visualization libraries help in analyzing results and performance metrics effectively. The flexibility of Python allows customization of models and parameters based on different CPS environments. Overall, Python provides a powerful and efficient platform for building intelligent, scalable, and reliable intrusion detection systems.

Python Features

Python Features include various functionalities in CPS Intrusion Detection System: –

- **Ease of Integration:** Python supports seamless integration of deep learning and data processing libraries such as TensorFlow, PyTorch, NumPy, and Pandas, enabling efficient development of intrusion detection models.
- **Parallel Processing:** Python enables faster data processing and model training using parallel computation techniques, which helps in handling large-scale CPS network traffic efficiently.
- **Exception Handling:** Built-in exception handling ensures smooth execution by managing errors during data preprocessing, model training, and prediction phases.
- **Readability and Maintainability:** Python's simple and clean syntax makes the code easy to understand, modify, and maintain, especially with modular design and reusable components.
- **Dynamic Typing:** Python allows flexible data handling without strict type declarations, making it easier to process diverse CPS network traffic data formats.
- **Community Support:** Python has a strong community that provides extensive documentation, libraries, and support for deep learning and cybersecurity applications.

5.2 TOOLS AND TECHNOLOGIES USED

5.2.1 UML Tools:

- **Star UML:** Star UML is a software modeling tool used to design system architecture and diagrams such as use case, class, sequence, and activity diagrams for the CPS intrusion detection system.

5.2.2 Programming Languages:

- **Python 3:** Python3 is the core programming language used in this project due to its simplicity, flexibility, and strong support for machine learning and deep learning applications. It is used for data preprocessing, feature extraction, model building, and intrusion detection in CPS environments. Python enables implementation of hybrid CNN-

LSTM models and GAN-based data augmentation using libraries like TensorFlow or PyTorch. It also supports handling high-dimensional network traffic data efficiently. Its extensive ecosystem makes it ideal for developing real-time cybersecurity solutions.

5.2.3 Programming Frameworks:

- **TensorFlow/PyTorch:** These are powerful deep learning frameworks used to build and train the CNN-LSTM model and GAN architecture for intrusion detection and data augmentation. They provide efficient computation and GPU support for large-scale CPS datasets.
- **Scikit-learn:** Used for preprocessing, feature scaling, dataset splitting, and evaluation metrics such as accuracy, precision, recall, and F1-score.

5.2.4 IDE:

- **Thonny:** A simple Python IDE useful for beginners to write and test basic code for data preprocessing and model implementation.
- **Visual Studio Code:** A widely used code editor with advanced features like debugging, extensions, and version control, suitable for developing and testing deep learning models

5.2.5 Libraries Of Packages:

In this project, Python and several open-source libraries are used for developing the CPS intrusion detection system:

1. **NumPy:** Used for numerical computations and handling large-scale network traffic data arrays.
2. **Pandas:** Used for dataset loading, preprocessing, and data manipulation.
3. **TensorFlow / PyTorch:** Used for implementing CNN-LSTM models and GAN-based data augmentation.
4. **Scikit-learn:** Used for data preprocessing, model evaluation, and performance metrics calculation.
5. **Matplotlib:** Used for visualizing training results, graphs, and performance metrics.
6. **Seaborn:** Used for advanced visualization such as heatmaps and confusion matrices.
7. **Gradient-based Explainable AI:** Used TensorFlow GradientTape to compute feature importance by analyzing the sensitivity of model outputs with respect to input features, enabling interpretation of CNN-LSTM predictions.

8. **OS:** Used for handling file operations such as dataset loading and saving model outputs.
9. **Sys:** Used for system-level operations and error handling.

5.3 ALGORITHM DETAILS

5.3.1 DATASET DETAILS

For the CPS intrusion detection system, the dataset used is the **Edge-IIoT Balanced Subset for Intrusion Detection dataset**, which contains realistic network traffic data collected from Industrial Internet of Things environments.

Selected Input Features:

S.No	Feature Name	Description
1	Frame Time	Represents the timestamp of each network packet
2	IP Source	Source IP address of the packet
3	IP Destination	Destination IP address of the packet
4	Source Port	Port number from which the packet is sent
5	Destination Port	Port number at which the packet is received
6	Flow Duration	Total duration of the network flow
7	Packet Rate	Number of packets transmitted per second
8	Protocol	Type of communication protocol (TCP/UDP)
9	Source Bytes	Amount of data sent from the source
10	Destination Bytes	Amount of data received at the destination
11	Payload Size	Size of the actual data carried in the packet
12	Total Packets	Total number of packets in a flow
13	Average Packet Size	Mean size of packets in the flow

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Table 5.3.1: Selected Input Features and its Description

- Name: Real-Time: Edgo-IIoT Balanced Subset for Intursion Detection Dataset
- Source: Publicly available cybersecurity dataset designed for IIoT and CPS environments.
- Size: Approximately 50,000 network traffic samples
- Traffic Variability: Includes both normal traffic and multiple attack types such as DDoS, DoS, MITM, and Injection attacks

- Annotations: Each record is labeled as either normal or a specific type of cyberattack.
- Format: Structured dataset in CSV format suitable for machine learning and deep learning models.
- Data Preprocessing: Includes normalization, encoding, feature extraction, and noise removal to improve model performance.
- Train-Validation-Test Split: Dataset is divided into training, validation, and testing sets for model training and evaluation.
- Security Considerations: Dataset is anonymized to ensure no sensitive or private information is exposed.

5.3.2 MODEL DETAILS

The proposed intrusion detection system for Cyber-Physical Systems is divided into several functional modules. Each module performs a specific task, and together they ensure efficient processing, analysis, and detection of cyber threats. The modular design improves system scalability, maintainability, and performance.

Data Collection Module

This module is responsible for collecting the dataset required for training and testing the intrusion detection system. The dataset used in this project is the Edge-IIoT dataset, which contains both normal and malicious network traffic.

The collected data includes multiple features such as packet size, protocol type, flow duration, and other network attributes. It also contains labeled data indicating whether the traffic is normal or represents a specific type of cyberattack.

This module ensures that the system has sufficient and relevant data for building an effective detection model.

Data Preprocessing Module

The raw dataset often contains noise, missing values, and irrelevant features. Therefore, preprocessing is necessary to improve data quality and model performance.

Key Steps in Preprocessing:

- **Data Cleaning:**

Removal of missing or inconsistent values.

- **Normalization:**

Scaling data into a standard range using Min-Max normalization:

$X' = (X - Xmin) / (Xmax - Xmin)$

This ensures that all features contribute equally to the model.

- **Encoding:**

Conversion of categorical values into numerical form.

- **Feature Selection:**

Selecting important features to reduce dimensionality and improve efficiency.

This module plays a crucial role in preparing the data for deep learning models.

GAN-Based Data Augmentation Module

One major challenge in intrusion detection is the imbalance between normal and attack data. This module uses Generative Adversarial Networks (GAN) to generate synthetic data samples.

Working of GAN:

GAN consists of two neural networks:

- **Generator (G):** Generates fake data samples
- **Discriminator (D):** Distinguishes between real and fake data

The objective function is:

$\min_G \max_D V(D,G) = E_{x \sim p_{data}(x)} [\log D(x)] + E_{z \sim p_z(z)} [\log(1 - D(G(z)))]$

This process helps in generating realistic synthetic samples, improving dataset balance and model performance.

CNN-LSTM Model Module

This is the core module of the system, responsible for detecting cyberattacks using deep learning.

CNN (Convolutional Neural Network):

CNN extracts spatial features from the input data using convolution operations.

Feature Map = Input * Kernel

It helps in identifying patterns and relationships between features.

LSTM (Long Short-Term Memory):

LSTM is used to capture temporal dependencies in sequential data.

Key equations:

Forget gate : $f_t = \sigma(W_f [h_{t-1}, x_t] + b_f)$

Input gate : $i_t = \sigma(W_i [h_{t-1}, x_t] + b_i)$

Cell State Update : $C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$

Output gate : $o_t = \sigma(W_o [h_{t-1}, x_t] + b_o)$

Hidden State : $h_t = o_t * \tanh(C_t)$

Activation Functions:

1) Sigmoid Function : $\sigma(x) = 1 / (1 + e^{-x})$
2) Tanh function : $\tanh(x) = (e^x - e^{-x}) / (e^x + e^{-x})$

LSTM helps in detecting time-based attack patterns.

Hybrid Model:

The combination of CNN and LSTM allows the system to:

- Capture spatial features (CNN)
- Capture temporal features (LSTM)

This improves overall detection accuracy.

Intrusion Detection Module

This module uses the trained CNN-LSTM model to classify incoming network traffic.

Functionality:

- Accepts new network data--
- Processes it through the trained model
- Classifies data as:
  - Normal
  - Attack (DDoS, DoS, MITM, Injection)

Output:

Prediction=f(Input Data)

It also generates alerts when malicious activity is detected.

Explainable AI (XAI) Module

Deep learning models are often considered black boxes. This module improves interpretability using gradient-based techniques. XAI is used to interpret the decisions made by the deep learning model. It identifies the most important features influencing intrusion detection results. XAI improves transparency, trust, and understanding of the model's predictions in CPS security systems.

Working:

- Uses TensorFlow GradientTape
- Computes feature importance using gradients:

Importance= ∂Output / ∂Input

This helps identify which features influence predictions the most.

Result and Visualization Module

This module presents the output of the system in an understandable format.

Functions:

- Display classification results
- Generate confusion matrix
- Plot accuracy and loss graphs

Performance Metrics:

- **Accuracy:**

Accuracy = (TP+TN) / (TP+TN+FP+FN)

- **Precision:**

Precision = TP / (TP+FP)

- **Recall:**

Recall = TP / (TP+FN)

- **F1-Score:**

F1 = (2×Precision×Recall) / (Precision+Recall)

These metrics help evaluate the performance of the intrusion detection system.

PYTHON MODULES

Python Python modules used in the CPS intrusion detection system include:

1. **NumPy:** Used for numerical computations and handling large datasets.
2. **Pandas:** Used for dataset loading, preprocessing, and data manipulation.
3. **Tensor Flow/PyTorch:** Used for implementing CNN-LSTM and GAN models.
4. **Scikit-Learn:** Used for preprocessing, dataset splitting, and evaluation metrics.

5. **Matplotlib:** Used for plotting graphs and performance visualization.
6. **Seaborn:** Used for advanced data visualization like heatmaps.
7. **Gradient-based Explainable AI (TensorFlow GradientTape)**: Used to interpret model predictions by computing feature importance based on the sensitivity of outputs with respect to input features.
8. **OS:** Used for file handling and system operations.
9. **Sys:** Used for system-level operations and error handling.
10. **Time / DateTime:** Used for tracking execution time and logging.

CHAPTER 6

SYSTEM TESTING

6.1 Introduction to System Testing

Testing strategies are used to evaluate the correctness, performance, and reliability of the proposed intrusion detection system. In this project, multiple testing approaches are applied to ensure that each module functions properly and the system as a whole performs efficiently in detecting cyberattacks in Cyber-Physical Systems.

System testing for CPS intrusion detection system involves:

- Verifying Verifying the functionality of intrusion detection, ensuring accurate classification of network traffic into normal and attack categories.
- Assessing communication between modules such as preprocessing, GAN, CNN-LSTM, and XAI for seamless integration.
- Evaluating system performance, response time, and resource utilization under varying traffic loads.
- Identifying and mitigating security vulnerabilities to protect CPS environments.
- Testing usability of the dashboard interface for monitoring, training, and detection.

6.2 Functional Testing:

Functional testing verifies that each function of the software application operates in conformance with the requirement specification. In the context of the CPS intrusion detection system, it ensures that all modules such as data preprocessing, GAN-based data augmentation, CNN-LSTM model training, intrusion detection, and XAI explanation function correctly and produce accurate results:

- **Unit Testing:** Each module such as preprocessing, GAN, CNN-LSTM model, and XAI module is tested individually to ensure correct functionality.
- **Integration Testing:** Testing integration between modules ensures smooth data flow from preprocessing → augmentation → model training → detection → explanation.
- **System Testing:** Validates complete workflow from dataset upload to intrusion detection and result visualization.

- **Interface Testing:** Tests dashboard features such as dataset upload, training, detection, and analytics display.

6.3 Non - Functional Testing:

Non-functional testing evaluates aspects of the CPS intrusion detection system that are not related to specific functionalities but focus on quality attributes such as performance, scalability, usability, and reliability. It ensures that the system efficiently handles large-scale CPS network traffic, provides quick response times for real-time detection, maintains compatibility across different platforms, and offers a user-friendly interface for monitoring and analysis.

- **Performance Testing:** Evaluates accuracy, response time, and system efficiency during real-time intrusion detection.
- **Load Testing:** Tests system behavior under large-scale CPS network traffic.
- **Compatibility Testing**: Ensures system works across different browsers and platforms.
- **Usability Testing:** Evaluates ease of use of dashboard and visualization outputs.
- **Security Testing:** Ensures secure handling of CPS data and protection from vulnerabilities.

6.4 Testing and Results:

Case id	Test Case Description	Expected Result	Actual Result	Status
1	Upload invalid dataset format (e.g., text file instead of CSV)	Error message: Only CSV dataset allowed	Error displayed correctly	Pass
2	Upload Edge-IIoT dataset	Dataset uploaded successfully	Dataset loaded successfully	Pass
3	Start preprocessing without dataset	Error message: Dataset required	Error displayed correctly	Pass
4	Perform data preprocessing	Data cleaned and normalized	Preprocessing completed successfully	Pass
5	Apply GAN augmentation on imbalanced data	Synthetic samples generated	GAN augmentation successful	Pass
6	Train CNN-LSTM model	Model training completed with high accuracy	Training completed successfully	Pass
7	Run intrusion detection	Traffic classified as normal/attack	Detection results displayed correctly	Pass
8	Evaluate performance metrics	Accuracy,Precision, Recall,F1-score	Metrics displayed correctly (~96%)	Pass
9	Generate XAI explanation	Important features identified	Feature importance displayed correctly	Pass
10	Test dashboard interface	All modules accessible and responding	UI works properly	Pass

Table 6.4: Testing and Results

6.5 Performance Testing

Performance testing is carried out to evaluate the efficiency, speed, and scalability of the proposed intrusion detection system. It ensures that the system performs well under different conditions and handles large volumes of data effectively.

The **processing time** of the system is measured to analyze how quickly it can preprocess data, train the CNN-LSTM model, and detect intrusions. Faster processing time indicates better system efficiency and suitability for real-time applications.

The **accuracy of the system** is evaluated using performance metrics such as accuracy, precision, recall, and F1-score. These metrics help in measuring how effectively the system detects cyberattacks while minimizing false positives and false negatives.

The **resource utilization** of the system is also analyzed by monitoring CPU and memory usage during execution. Efficient utilization ensures that the system can run smoothly without excessive hardware requirements.

Scalability testing is performed by increasing the size of the dataset to evaluate how well the system handles large-scale CPS environments. The system should maintain performance even with increased data volume.

The **real-time performance** of the system is tested to ensure that it can detect intrusions with minimal delay. This is important for practical deployment in CPS environments where immediate response is required.

Overall, performance testing confirms that the system is efficient, scalable, and capable of providing accurate intrusion detection in real-time scenarios.

6.6 Results and Output Screens:

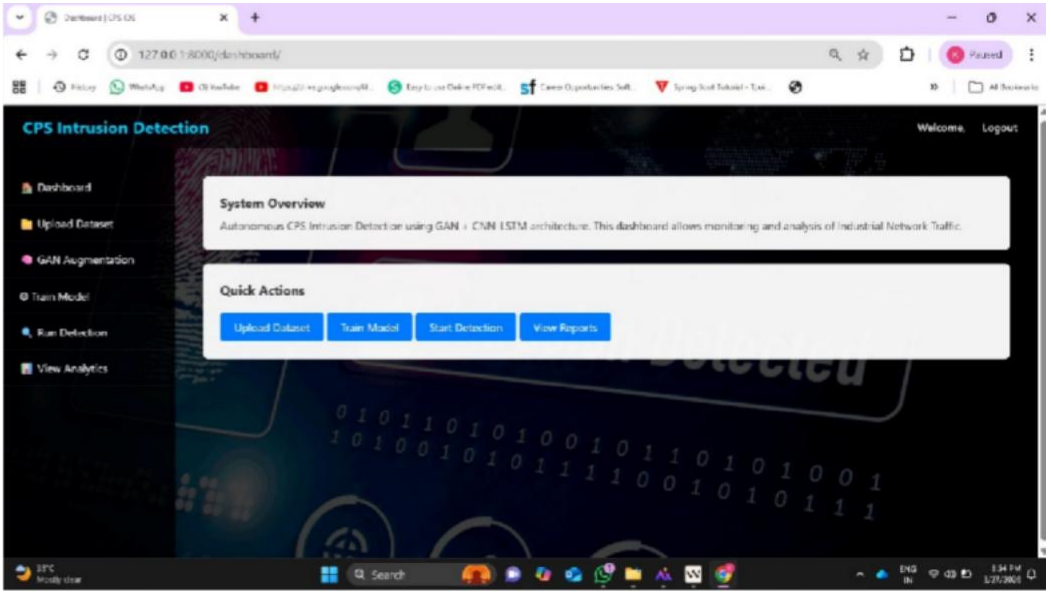
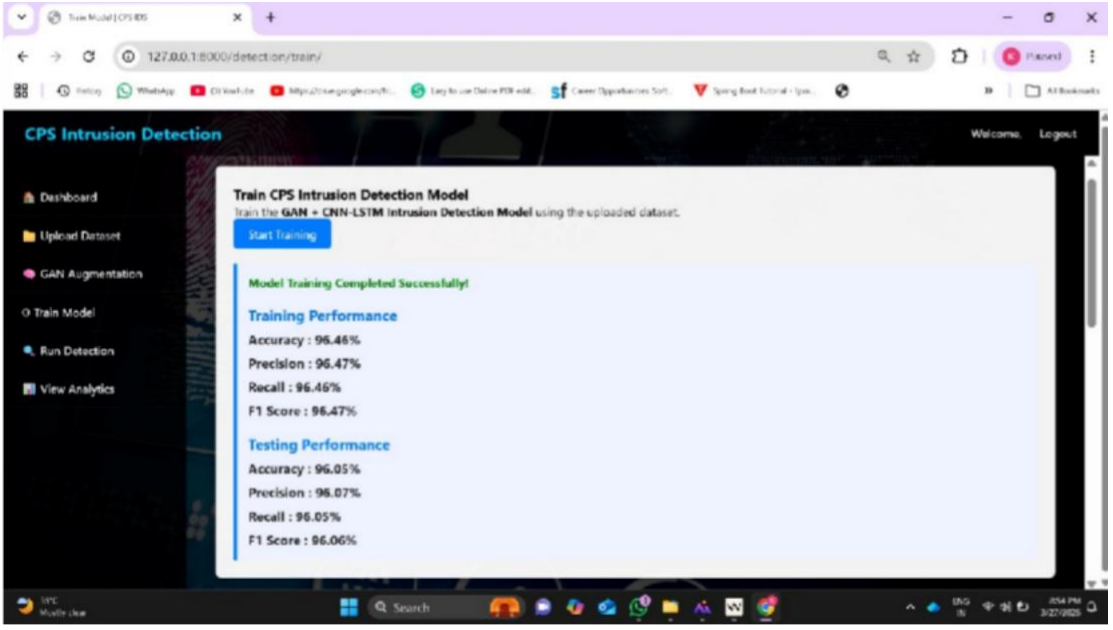


Figure 7: Displayed CPS Intrusion Detection System Dashboard Interface.

This figure presents the main dashboard of the Cyber-Physical System (CPS) Intrusion Detection System. The interface provides a centralized view of system activities, including real-time network traffic monitoring, alert notifications, and system status indicators. It enables users to quickly assess the health of the network and identify potential security threats through visual summaries and interactive components.



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Figure 8: Displayed Model Training Results and Performance Metrics.

This figure illustrates the training results of the machine learning model used for intrusion detection. It includes key performance metrics such as accuracy, precision, recall, and F1-score. The visualization helps evaluate how effectively the model learns patterns from the dataset and its capability to distinguish between normal and malicious activities.



Figure 9: Displayed Intrusion Detection Output with XAI Feature Importance.

This figure shows the output of the intrusion detection system along with Explainable AI (XAI) insights. It highlights the most influential features contributing to the model's decision-making process. By providing feature importance, the system enhances transparency and helps users understand why a particular network activity is classified as normal or anomalous.

Additionally, this figure demonstrates how explainability enhances the usability of machine learning in critical CPS environments, where understanding the reasoning behind predictions is as important as the predictions themselves. It supports better decision-making, faster incident response, and improved system reliability.

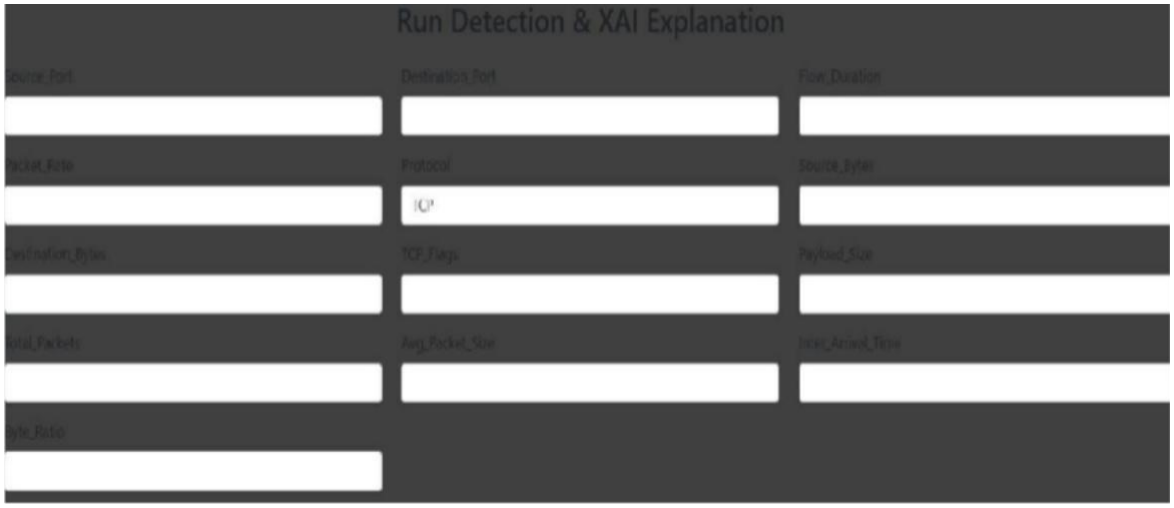


Figure 10: Real-Time Detection Input Interface for CPS Network Traffic.

This figure presents the real-time input interface used to feed network traffic data into the CPS intrusion detection system. The interface is designed to accept various network parameters such as source and destination IP addresses, port numbers, protocol types, packet sizes, and timestamps. It allows users to either manually input data or stream live network traffic for continuous monitoring.

The system processes the input instantly and forwards it to the trained model for classification. This enables real-time detection of potential intrusions, making the system suitable for deployment in dynamic and time-sensitive CPS environments. The interface may also include validation mechanisms to ensure data accuracy and prevent incorrect inputs.

Furthermore, the interface supports user interaction by displaying immediate results, such as classification output (normal or attack) and confidence scores. This real-time feedback loop helps users quickly identify suspicious activities and take preventive measures. The design emphasizes usability, responsiveness, and seamless integration with backend detection modules, ensuring efficient and reliable operation in practical scenarios.

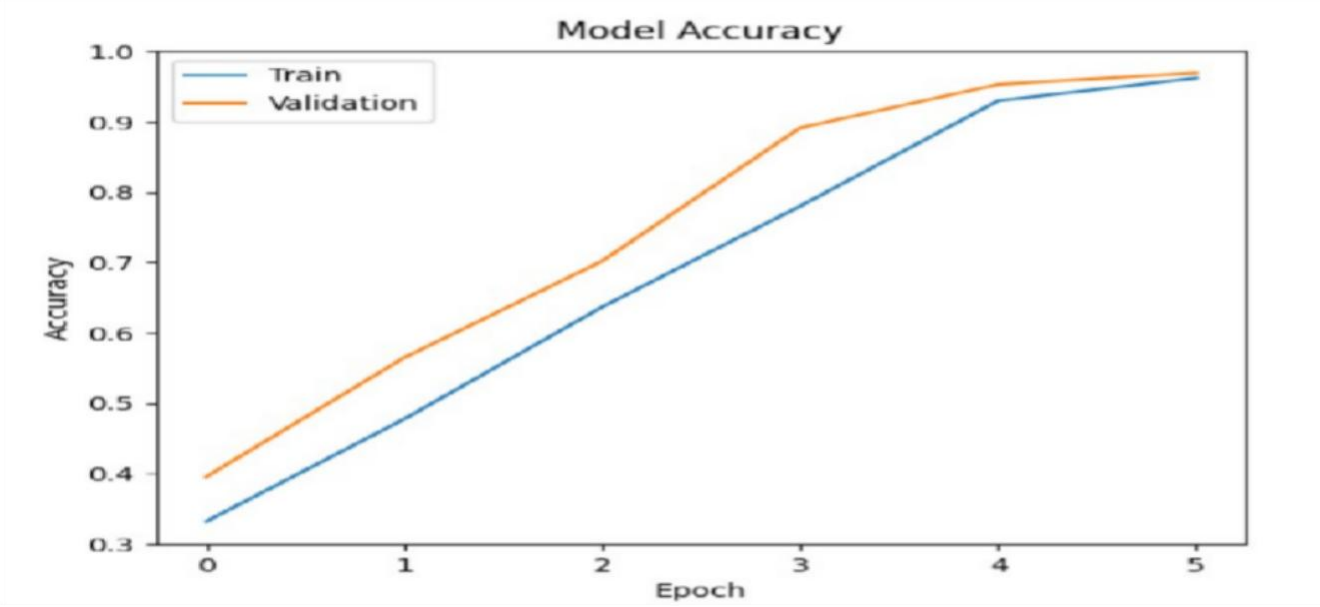


Figure 11: Accuracy Graph

This figure represents the accuracy of the model over training epochs. It shows how the model's performance improves as it learns from the data. A steady increase in accuracy indicates effective learning, while stabilization suggests convergence of the model.

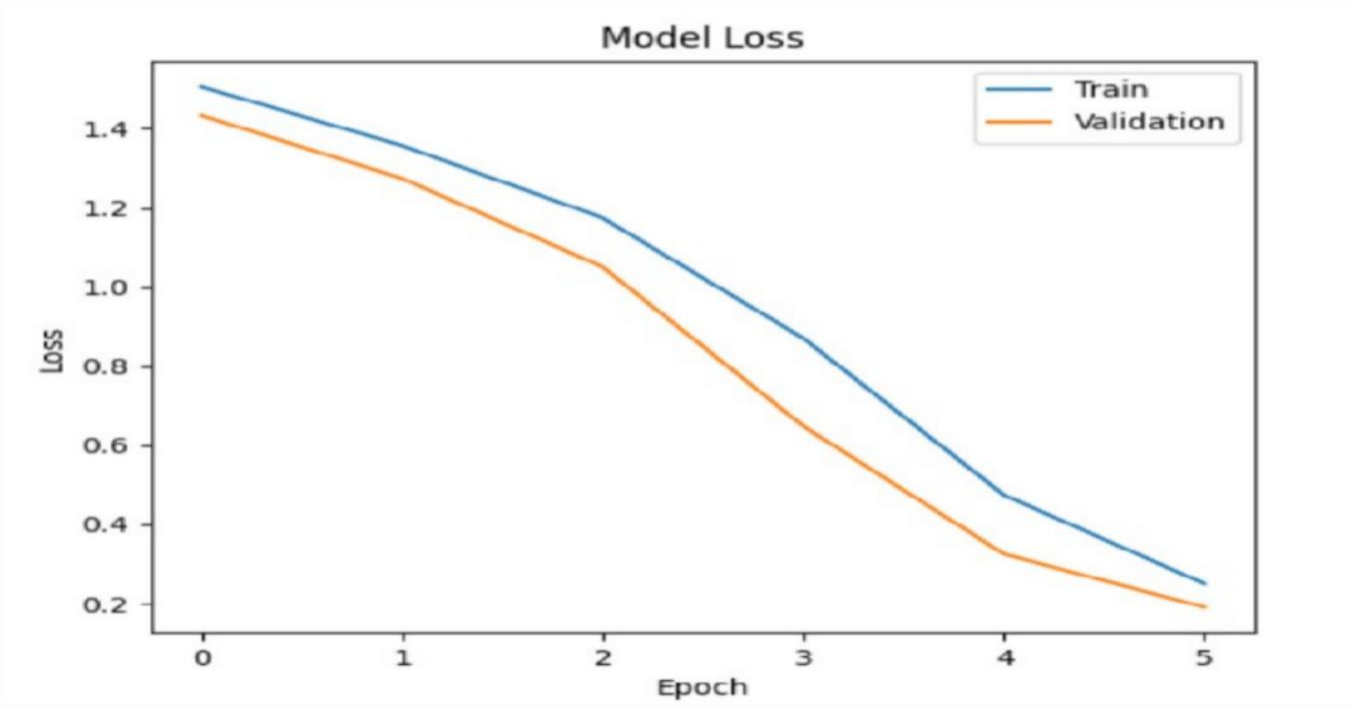


Figure 12: Loss Graph

This figure illustrates the loss value of the model during training. The decreasing trend in loss indicates that the model is minimizing errors and improving prediction performance. It is useful for identifying issues such as underfitting or overfitting.

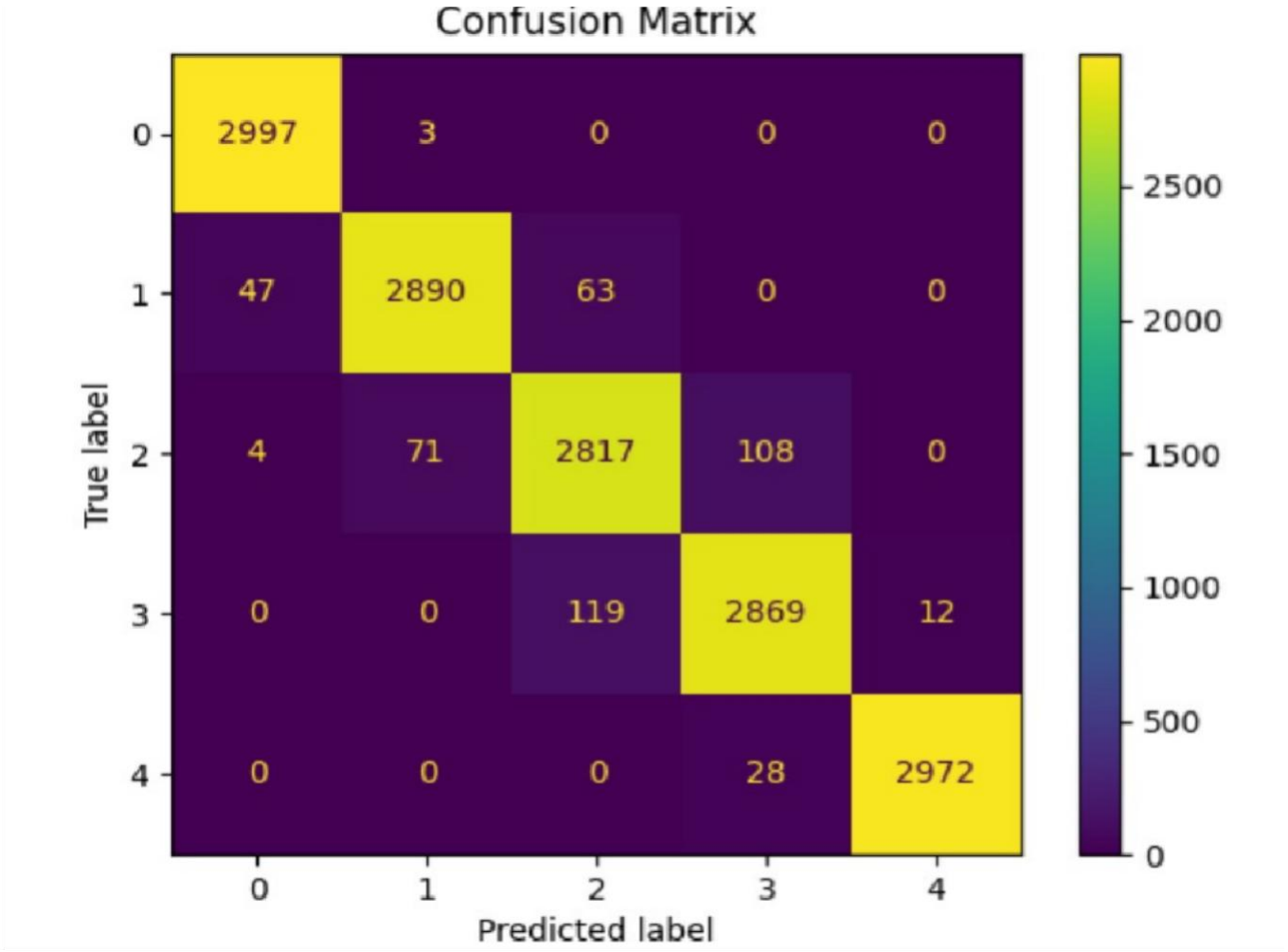


Figure 13: Confusion Matrix

This figure displays the confusion matrix used to evaluate the classification performance of the intrusion detection model. It shows the distribution of true positives, true negatives, false positives, and false negatives. This helps in understanding how accurately the model classifies different types of network activities and where misclassifications occur.

6.7 Discussions

The results obtained from the system show that the proposed approach is effective in detecting cyber threats in CPS environments. The hybrid CNN-LSTM model plays a significant role in improving detection accuracy by capturing both spatial and temporal dependencies in the data.

The use of GAN-based data augmentation addresses the problem of imbalanced datasets, which is a common issue in intrusion detection systems. By generating synthetic data, the system becomes more robust and capable of detecting rare attack patterns.

The integration of Explainable Artificial Intelligence provides transparency in model predictions, allowing users to understand which features influence the detection process. This improves trust and usability of the system.

The system also demonstrates good real-time performance with efficient processing and low response time. It is scalable and can handle large CPS datasets without significant performance degradation.

However, there are some limitations. The system is tested on a dataset and not deployed in a real-world CPS environment. The performance may vary with different datasets and real-time conditions. Future improvements can include real-time deployment, optimization of model performance, and extension to detect more advanced cyber threats.

6.8 Summary

The results and discussion confirm that the proposed intrusion detection system is effective, accurate, and reliable. The combination of CNN-LSTM, GAN, and Explainable AI techniques enhances the system's performance and makes it suitable for modern CPS security applications.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

The proposed intrusion detection system for Cyber-Physical Systems (CPS) has been successfully designed and implemented using advanced deep learning techniques. The system utilizes a hybrid CNN-LSTM model to effectively capture both spatial and temporal features of network traffic, resulting in improved detection accuracy.

The integration of GAN-based data augmentation helps in addressing the issue of imbalanced datasets, thereby enhancing the system's ability to detect rare and complex cyberattacks. Additionally, the use of Explainable Artificial Intelligence (XAI) provides transparency in model predictions, making the system more reliable and trustworthy.

The system is capable of accurately classifying network traffic into normal and malicious categories and successfully detects various types of cyberattacks such as DDoS, DoS, MITM, and injection attacks. Performance evaluation results demonstrate that the system achieves high accuracy with minimal false positives and false negatives.

The results obtained demonstrate that the system performs efficiently in identifying multiple cyberattacks such as DDoS, DoS, MITM, and injection attacks with high accuracy and low error rates. The project contributes to improving the security and reliability of CPS environments, making it suitable for modern industrial applications.

Overall, the project presents an efficient, scalable, and intelligent solution for securing CPS environments against modern cyber threats.

7.2 FUTURE SCOPE

- Future work can focus on integrating advanced AI techniques such as Transformer models and reinforcement learning to further improve intrusion detection accuracy in CPS environments, while expanding the system for real-time deployment using edge computing and cloud integration to enhance scalability and performance. Incorporating automated

response mechanisms can enable immediate action against detected cyber threats, improving system resilience. Enhancing Explainable Artificial Intelligence (XAI) techniques can provide better visualization and interpretability for faster decision-making by security analysts. Integration with blockchain technology can improve data integrity and secure communication in CPS networks. Additionally, continuous dataset expansion and training with real-world attack scenarios can strengthen the system's ability to detect zero-day and evolving cyber threats effectively.

The system can be extended for real-time deployment in actual industrial environments to improve practical applicability. Future work can also focus on optimizing the model to reduce computational complexity and improve processing speed.

Additional types of cyberattacks can be included to make the system more comprehensive. The integration of advanced deep learning techniques and hybrid models can further enhance detection performance.

Furthermore, the system can be deployed using cloud or edge computing platforms to improve scalability and accessibility. Enhancing the Explainable AI component for better interpretability and user understanding can also be considered.

Overall, future enhancements can make the system more robust, efficient, and suitable for large-scale real-world CPS applications.

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Hybrid Deep Learning-Based Intrusion Detection for Cyber-Physical Systems with Explainability

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Abstract

Cyber-Physical Systems (CPS) are increasingly vulnerable to cyberattacks due to their interconnected nature in Industrial Internet of Things (IIoT) environments. Detecting such attacks is challenging due to high-dimensional and complex network traffic data. This study proposes a hybrid deep learning-based intrusion detection system combining Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), and Generative Adversarial Networks (GAN) for enhanced detection performance. The model is evaluated using the Edge-IIoT Balanced Subset dataset containing 50,000 samples across multiple attack categories. After preprocessing, a total of 14 attributes were retained from the dataset. Out of these, 13 were used as input features for model training, while the remaining attribute (Attack_Label) was used as the target variable. Features such as Frame_Time, Source_Port, and Destination_Port were excluded as they do not significantly contribute to intrusion detection and may introduce noise into the model. CNN is used for spatial feature extraction, LSTM captures temporal dependencies, and GAN improves data balance through synthetic sample generation. Additionally, Explainable Artificial Intelligence (XAI) techniques are applied to interpret model decisions. Experimental results show that the proposed model achieves an accuracy of 96.2%, demonstrating its effectiveness in detecting cyberattacks while maintaining interpretability. This approach provides a reliable and scalable solution for CPS security.

Keywords—Cyber-Physical Systems, Intrusion Detection System, CNN- LSTM, Generative Adversarial Networks, Explainable Artificial Intelligence, Network Traffic Analysis, Attack Detection, Edge- IIoT Balanced Subset for Intrusion Detection Dataset.

I. INTRODUCTION

The rapid expansion of intelligent industrial infrastructure has led to increased usage of Cyber-Physical Systems (CPS) across numerous modern applications, including Smart Manufacturing, Intelligent, Smart Grids and robots [5], [7]. CPS combines computational intelligence, communication networks, and physical processes to monitor in real-time, exchange data, and provide automated control over physical environments via the Industrial Internet of Things (IIoT) [10], [22]. While this level of integration provides enhanced operational efficiency and automation, the level of interconnectivity within CPS also creates new, unique challenges with regard to cybersecurity [6], [7]. Due to their interconnectivity, CPS environments have become primary targets for sophisticated cyber-attacks that target various vulnerabilities in both network communication and the interaction of devices with each other [5], [20].

Traditional intrusion detection systems typically rely on a combination of signature or rule-based mechanisms for detection; thus, these mechanisms are not designed to detect attacks that have evolved over time and are complex when looking across multiple CPS networks [14], [17]. Also, the volume of high-dimensional network traffic generated by CPS environments makes the task of detecting attacks significantly more difficult using conventional machine learning methods [18].

In recent years, deep learning techniques have shown great promise for use in cybersecurity applications because they automatically extract complicated patterns from very large datasets [13], [19]. To address this problem, we proposed a hybrid CNN-LSTM based intrusion detection system, along with an enhanced version that incorporated GAN (Generative Adversarial Network) based data augmentation techniques, as well as a hybrid explainable AI (XAI) model that is able to produce interpretable results. We performed the evaluation for the newly developed hybrid models using the Edge-IIoT Balanced Subset for Intrusion Detection dataset. It contains 50,000 samples for five traffic classes (DDoS, DoS, MITM, Injection and normal traffic) and an improvement in intrusion detection accuracy, while providing interpretable outputs of the top features that influence the security of CPS networks.

Despite significant advancements in deep learning-based intrusion detection systems, several critical challenges remain unresolved. Most existing approaches focus primarily on improving detection accuracy while neglecting model interpretability, which is essential for decision-making in critical cyber-physical systems. Additionally, many methods fail to effectively handle class imbalance in network traffic data and do not integrate spatial, temporal, and data augmentation techniques within a unified framework. Furthermore, existing models often lack transparency, making it difficult for security analysts to trust automated decisions. These limitations highlight the need for a hybrid, explainable, and robust intrusion detection system for CPS environments. The main contribution of this work is the integration of CNN, LSTM, and GAN within a unified framework along with explainable AI techniques to improve both detection accuracy and model transparency in CPS environments.

II. LITERATURE REVIEW

A. CNN-LSTM Based Intrusion Detection

CNNs have become one of the most commonly used methods of feature extraction in cybersecurity applications [13], [14]. CNNs can effectively model the spatial relationships among input features and automatically learn hierarchical representations from network traffic data, thereby improving the classification accuracy of intrusion detection [14]. LSTMs are also used extensively for time series data analysis, including traffic flows [2], [19]. In network security applications, LSTMs have been successfully applied to detect temporal relationships in traffic flows, as well as to recognize abnormal behaviors that develop over time [2], [19]. LSTM models capture sequential dependencies, while CNN models capture spatial dependencies.

B. Generative Adversarial Network Based Augmentation

A major challenge faced by intrusion detection systems is class imbalance, where certain types of attacks have significantly fewer samples compared to the large amount of normal network traffic [1], [5]. This imbalance can negatively affect the performance of deep learning models. To address this problem, researchers have explored the use of Generative Adversarial Networks (GANs) for data augmentation [1], [5]. GANs consist of two competing neural networks—a generator and a discriminator—that work against each other during training to produce realistic synthetic data samples. In cybersecurity, GAN-based and similar models are employed to generate additional attack samples, helping to balance datasets and improve the robustness of intrusion detection systems [1], [5]. By augmenting the training data with these synthetic samples, GANs enable deep learning models to better represent rare attack patterns and learn them more effectively.

C. Explainable Artificial Intelligence (XAI) in cybersecurity

Many deep learning models have experienced great success when used in security but many of these deep learning models are essentially "black box" in nature and thus have a major limitation with regards to how the prediction was made (i.e., not easily interpretable) [13], [18]. As no explanation exists for the prediction made; it creates a negative impact on how much trust the user placed upon utilizing an Intrusion Detection System (IDS) that is automated, because the prediction lacked sufficient transparency. In addition, if users have little or no trust in utilizing an IDS - Automated, then they may not incorporate automated methods into critical Infrastructures (i.e., cyber/physical systems), thereby limiting the practical applicability [18]. These methods provide insight into how / why a model made a particular decision by providing an understanding of which features and/or internal components within the model contains the greatest impact or influence on the final classification outcome. Using explainability techniques in an Intrusion Detection (ID) system provides security analysts with an understanding of which characteristics are present in network traffic that help to identify malicious activity [13], [18]. Researchers have used methods such as feature importance analysis, LIME (layer-wise interpretation), and contribution analysis to determine the influence of different model features as well as how specific neural network layers impact the overall classification outcomes. Therefore, by applying explainable Artificial Intelligence (XAI) methods to deep learning models, Intrusion Detection Systems provide both detection performance and meaningful explanations of their predictions of detected incidents.

D. Comparative Analysis of existing approaches

With the increasing reliance on Cyber-Physical Systems, cybersecurity and intrusion detection have become major concerns [5], [7]. Researchers have explored the use of AI and deep learning to improve detection system performance, make systems more resilient to failures, and reduce cyber threats [13], [19]. Techniques like CNN, LSTM, and GAN have been studied extensively to enhance intrusion detection [1], [14], [19]. These approaches have proven to be much better at detecting complex and new attack patterns compared to traditional methods [14], [17]. Table 1 gives an overview of recent research in intrusion detection systems, covering different datasets, methods, key contributions, and limitations.

Ref	Author & Year	Technique Used	Dataset / Application Area	Key Strength
[2]	Levytskyi et al. (2024)	Deep Learning + Reinforcement Learning	Autonomous CPS	Improves adaptive decision-making
[3]	Divya et al. (2025)	Hybrid RL + Supervised Learning	CPS fault detection	Enhances fault tolerance
[5]	Sharma (2024)	CPS Security & Privacy Framework	Cyber-Physical Systems	Addresses security challenges
[6]	Phillips et al. (2024)	Knowledge-based Risk Assessment	CPS systems	Automated cybersecurity risk analysis
[7]	Perone et al. (2024)	Cybersecurity Risk Assessment	Autonomous CPS	Focus on safety and security
[13]	Shelke et al. (2025)	Federated Learning for CPS Security	CPS security datasets	Decentralized intrusion detection

Table 1: Comparison between existing techniques

The majority of the studies reviewed provide evidence that AI and Deep Learning have improved Intrusion Detection and cybersecurity performance [13], [19]. However, there continue to be several areas of concern that must be addressed relating to computational complexity, limited scalability of models, and lack of interpretability [6], [18], [21]. These areas of concern highlight the importance of developing hybrid systems which combine computationally efficient detection methods with greater transparency and explainability than currently available.

E. Research Gap

Despite extensive research on deep learning techniques such as CNNs, LSTMs, and GANs for intrusion detection in cyber-physical systems, most existing approaches primarily emphasize detection accuracy while overlooking critical aspects of model explainability and interpretability [8], [11], [27]. Several studies have explored deep learning–based intrusion detection frameworks and highlighted challenges such as data imbalance, spatiotemporal feature extraction, and model complexity; however, transparency in model decision-making remains insufficiently addressed [26], [28].

Although GANs have been investigated to mitigate issues such as class imbalance and the scarcity of attack samples, their integration with hybrid deep learning architectures is still limited [27], [17]. Furthermore, existing research in cyber-physical system security largely focuses on enhancing system robustness, risk assessment, and AI-driven threat detection, with comparatively less attention given to interpretable model design [5], [6], [13].

In addition, only a few studies adopt a bottom-up interpretability perspective, where model behavior is analyzed through feature contribution assessment and internal layer-wise understanding [18], [21]. This lack of interpretability reduces the transparency, trustworthiness, and practical deployment of intrusion detection systems in safety-critical cyber-physical environments [4], [7].

To address these gaps, this research proposes a hybrid deep learning-based intrusion detection system that integrates a CNN–LSTM architecture with GAN-based data augmentation, while explicitly incorporating explainability techniques. These include layer-wise interpretability analysis and feature importance assessment to identify the most influential features and internal network components contributing to final predictions. By doing so, the study aims to enhance both detection performance and the transparency of intrusion detection systems in cyber-physical systems.

III. METHODS

A. System Architecture

This proposed intrusion detection framework contains multiple sequential phases such as data preprocessing, feature extraction, classification, attack detection, and explanation [3], [15]. The Cyber-Physical Systems (CPS) are combination of physical components, cyber components, and communication components [5], [9].

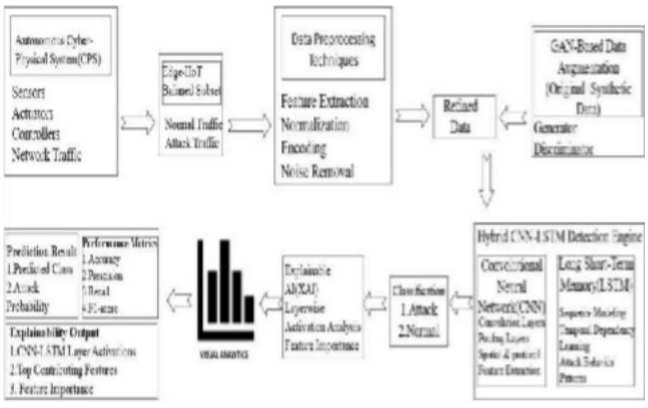


Fig. 1. System Architecture

Initially, network traffic data from the dataset was preprocessed to remove inconsistencies and normalize feature values. Next, preprocessed data was provided as input to a hybrid deep learning architecture with combined CNN and LSTM layers [3], [15]. The CNN portion contains CNN layers that perform the automatic extraction of features by identifying spatial relationships within input features, and the LSTMs learn the temporal dependencies associated with sequential network traffic flows [2], [19]. To enhance the overall robustness of the model, synthetic data samples were generated via a GAN during the training phase of the framework [1], [5]. XAI techniques were also used to analyze predicted attacks based on the model's predictions to determine which features were most influential in detecting cyberattacks [13], [18].

B. Dataset Description

The Edge-IIoT Balanced Subset dataset was selected due to its relevance to real-world Industrial Internet of Things (IIoT) environments. It contains approximately 50,000 network traffic samples, including both normal and malicious activities. The dataset includes multiple attack categories such as Distributed Denial of Service (DDoS), Denial of Service (DoS), Man-in-the-Middle (MITM), and Injection attacks.

Initially, the dataset contained 17 attributes. After preprocessing, 14 attributes were retained. Among these, 13 attributes were used as input features for model training, while one attribute (Attack_Label) was used as the target variable.

Features such as Frame_Time, Source_Port, and Destination_Port were removed during preprocessing, as they do not significantly contribute to intrusion detection and may introduce noise into the model. We used 70% of the dataset for training and 30% of the dataset for testing.

The balanced nature of the dataset ensures that the model does not suffer from bias toward any particular class, thereby improving detection performance and generalization.

C. Data Preprocessing

The first step is to clean the data by removing any rows with missing values or columns with irrelevant features to ensure we have quality data to work with. After cleaning and preparing your data, a feature scaling technique must be used to convert all numeric values into a common range (i.e., between 0 and 1). This allowed your neural networks to have more significant stability and converge more rapidly during training [18]. Feature selection helps in reducing model complexity, improving training efficiency, and enhancing detection accuracy by eliminating redundant and irrelevant attributes.

Normalize Equation:

$$X_{norm} = (X - X_{min}) / (X_{max} - X_{min})$$

Where;

X = original feature value

Xmin = minimum feature in the dataset

Xmax = maximum feature in the dataset

Xnorm = normalized value

D. CNN-LSTM Intrusion Detection Model

The primary part of the recommended framework is a combined CNN and LSTM deep learning network for recognizing cyberattacks based on information acquired from network traffic data [3], [15]. The CNN layer takes the input data and conducts convolution operations on all the feature maps within it to discover high-level characteristics associated with the nature of the threat or malicious activity in the input data [14].

The proposed CNN-LSTM architecture consists of multiple layers designed to effectively capture both spatial and temporal features of network traffic data. Initially, the input data is passed through

convolutional layers with multiple filters to extract spatial feature representations. These layers are followed by activation functions (ReLU) and pooling layers to reduce dimensionality and retain important features.

The extracted feature maps are then flattened and passed to the LSTM layer, which is responsible for learning temporal dependencies and sequential patterns in the data. The LSTM network contains memory cells that retain information over time, enabling the detection of complex attack patterns that evolve across network traffic sequences.

Finally, the output from the LSTM layer is passed through fully connected (dense) layers, followed by a softmax classifier to categorize the traffic as either normal or malicious. This hybrid architecture enables the model to achieve improved accuracy by combining both spatial and temporal learning capabilities.

To compute the convolution between input and kernel data, the following equation must be utilized:

$$F(i,j)=(I*K)(i,j)$$

Where,

– I= input data; K= convolution kernel and F(I,j) = output feature map.

The CNN component uses multiple convolutional layers with kernel size 3×3 and max-pooling operations, while the LSTM layer consists of 64 memory units to capture temporal dependencies.

Once network traffic features have been extracted via convolution, they may then be passed to an LSTM network for the discovery of sequential dependencies amongst them. The LSTM model is capable of determining the time-based (temporal) pattern of occurrences during a cyberattack [2], [19].

E. Data Augmentation Using GAN

The Generative Adversarial Networks (GAN) used to generate synthetic attack samples for data augmentation enhances the performance of an Intrusion Detection System (IDS) [1], [15]. GANs use of deep learning allows for the generation of new data that is substantially similar to existing data sets (real data) by combining two neural networks, the Generator and the Discriminator.

Generator creates synthetic (fake) samples (xfake) from random input (z):

$$xfake = G(z)$$

where (z) is a random input and G(z) is a synthetic sample that corresponds to (z).

The Discriminator (D) used to determine whether the sample (x) is real or fake: D(x)

where D(x) is the probability (between 0 and 1) that the input data sample (x) is real.

The objective function for GANs is defined as follows:

$$\min G \max D\ V(D, G) = E[\log D(x)] + E[\log(1 - D(G(z)))]$$ Where,

G = Generator, D = Discriminator, x = real data, z = random noise,

D(x) = amount of data that is real

G(z) = fake data generated from noise

F. *Explainable Artificial Intelligence(XAI)*

Deep learning models are commonly viewed as being operated by black-box systems which makes it difficult for people to interpret how they make decisions[13], [18]. To remedy this situation, XAI techniques were applied to the output of the model to determine how it arrived at its prediction. The analysis consists of three parts:

1. Layer Analysis - This accounts for the contribution of each layer in the neural network using the attack detection model,

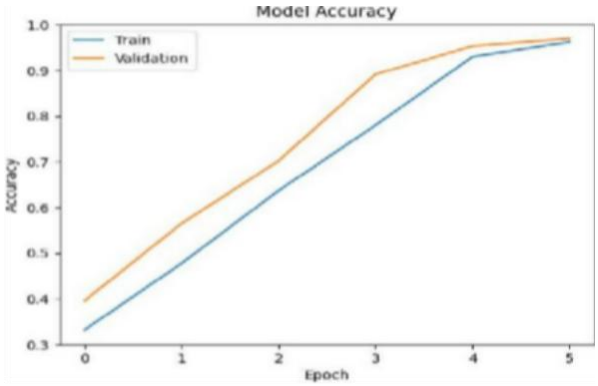
2. Feature Importance - This indicates the features in the network which had the biggest contribution on how the system makes its predictions,

3. Top Contributing Features - This indicates which features are most critical for identifying malicious network traffic.

IV. Results and Discussions

A. *Experimental Setup*

This study utilizes the Edge-IIoT Balanced Subset for Intrusion Detection dataset, which includes normal and malicious network traffic samples from Industrial Internet of Things (IIoT) environments and various types of cyberattacks and normal communication behavior. Prior to model training, the dataset was preprocessed to remove features that did not contribute to model prediction and to ensure that it was formatted appropriately for deep learning analysis [18]. The preprocessed dataset was then divided into training and testing datasets, the testing dataset contained the minority of records while the majority comprised the training dataset. The model was trained using deep learning during multiple epochs until the learning performance reached stability. During model training, the model learned patterns which were used to distinguish between normal network activity and malicious activity. Performance metrics of the system were evaluated with respect to traditional classification metrics that are utilized in research on intrusion detection systems through the use of the Edge-IIoT Balanced Subset for Intrusion Detection dataset [14], [17].



B. *Training Performance Analysis*

Fig. 2. Accuracy Graph

The above graph shown in Fig. 2. demonstrates how the proposed model's prediction capabilities improve with training. As training epochs are introduced, the model's prediction accuracy grows increases, indicating that the model is successfully extracting characteristics from the input data.

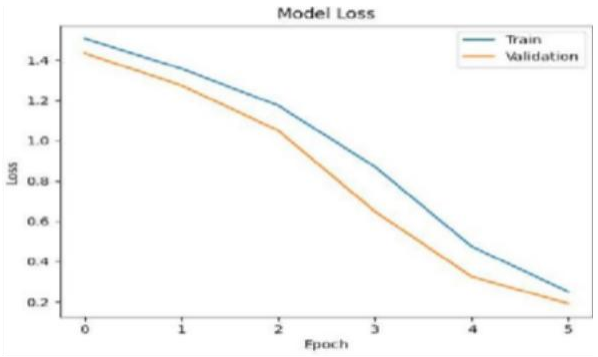


Fig. 3. Loss Graph

The above graph shown in Fig. 3. depicts errors between predicted outputs and actual labels during training. Since there was not an existing pattern of the input data at the beginning of training, the model produced a higher loss than at any later stage. As the model trains, the loss value should decrease with each subsequent epoch, indicating that the model has developed better prediction capabilities and reduced classification errors. Hence, the behavior of increased accuracy and decreased loss validates that the proposed model has learned characteristics of network traffic and improved its detection capabilities.

C. *Confusion Matrix Analysis*

A confusion matrix is an effective method of visualizing how well the classifier has classified instances into one of the five categories: "normal traffic," or "attack traffic". For the classification of each of the five classes, the matrix provides visually detailed information on both how many total samples were classified correctly and incorrectly into each category.

The diagonal elements of your confusion matrix correspond to the samples that were predicted correctly, meaning both "normal" and "attack" classified instances are shown as correctly classified instances for each attack type.

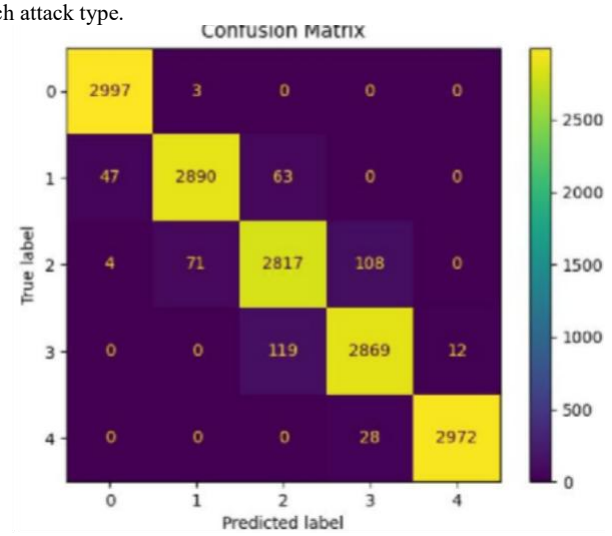


Fig. 4. Confusion Matrix

Similarly, the off-diagonal elements correspond to the instances of each attack type that were misclassified. A "good" performing intrusion detection system had a majority of the numbers in each category consisting primarily of diagonal elements (poorly categorized predictions consist primarily of off-diagonal elements).

D. *Explainability Analysis Using XAI*

XAI Techniques that Improve transparency and trust in an Intrusion Detection System applied for transparency and trust in an intrusion detection system, Explainable Artificial Intelligence (XAI) methods are used to analyze the prediction process of the deep learning model [13], [18].

1. Layer-wise Analysis - The layer-wise analysis provides insight regarding how the different layers of the neural network are utilized to process the input to the system. This structure of learning in layers gives the neural network the ability to learn and provide insights regarding very fine differences in the behaviors of normal and malicious actors.
2. Feature Importance - The feature importance analysis provides insight regarding the most important features of the input used to obtain prediction results. In some cases, certain network traffic features are found to have greater influence on the detection process, thereby identifying which features are most significant for identifying abnormal patterns [13], [18].
3. Top Contributing Features - The top contributing features highlighted during the analysis indicate the features that have the largest impact on the detection of an intrusion. That is, these features would provide valuable insight into which characteristics of network traffic are most helpful in discriminating between attacks and normal network communications.

E. *Final Results Summary*

The proposed intrusion detection model was tested using key classification metrics or performance metrics to assess its effectiveness in distinguishing between normal and harmful network traffic. These measurements offer a clear picture of how well the model performs in identifying complex patterns within cyber-physical systems.

Formulas:

Accuracy = (TP+TN) / (TP+TN +FP+ FN)

Precision = (TP) / (TP + FP Recall

= (TP) / (TP + FN)

F1-score = (2 × (Precision ×Recall)) /(Precision + Recall) Where,

TP = True Positive, TN = True Negative, FP = False Positive, FN = False Negative.

Model	Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
CNN Model	CPS Dataset	89.5	88.7	89	88.8
LSTM Model	CPS Dataset	91.2	90.5	91	90.7
CNN-LSTM Hybrid	Edge-IIoT	94.3	93.8	94.1	93.9
Proposed Model (CNN-LSTM + GAN + XAI)	Edge-IIoT	96.2	95.8	96.5	96.1

Table 2. Results Summary

Table 2 shows that the model consistently performs well across all metrics. It achieves 96.2% accuracy, indicating that it can generally classify input data correctly. With a precision of 95.8%, the model effectively reduces false positives. The recall rate of 96.5% demonstrates its strong capability in detecting true attack instances. The F1-score stands at 96.1%, signifying a good balance between precision and recall. These findings suggest that the model is both dependable and effective for use in intrusion detection within cyber-physical environments.

F. *Discussions*

The information obtained demonstrates that our model is able to find intrusion pattern activity within the Edge-IIoT Balanced Subset for Intrusion Detection dataset with high reliability in classifying both normal and malicious traffic, as demonstrated by the majority of prediction classes within the confusion matrix being accurate [14], [17]. This supports the model's ability to learn typical traffic patterns associated with IIoT-based cyberattacks [5], [20].

Using XAI analysis also helped to further understand the model by revealing what features contributed the most to influence the predicted outcome [13], [18]. Thus, our proposed method is both effective for identifying intrusions and more interpretable for use in cybersecurity applications than traditional approaches [13], [19]. Overall, this indicates that our proposed model can be effectively used for intrusion detection in both cyber-physical and IIoT environments [5], [7].

G. **Conclusion and Future Work**

This paper proposed a hybrid deep learning-based intrusion detection system for Cyber-Physical Systems using CNN, LSTM, and GAN with explainable AI techniques. The proposed model effectively captures both spatial and temporal patterns in network traffic data and improves detection accuracy. Experimental results on the Edge-IIoT dataset demonstrate that the model achieves high accuracy (96.2%) and provides interpretable insights into cyberattack detection. The integration of XAI techniques enhances transparency and helps in understanding the decision-making process of the model. This enhances the reliability and practical applicability of the system in real-world Cyber-Physical System environments. In future work, the model can be extended to handle real-time intrusion detection and tested on more diverse datasets. Additionally, optimization techniques can be applied to reduce computational complexity and improve scalability.

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